

# Cartographic Visualization

## 1<sup>st</sup> Exercise

Subject: “Cartographic visualization using different types of static maps”

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### Description

This work is based on the construction of some cartographic products, as we will see below. Our aim is to create these static maps by selecting a prefecture of Greece.

### Data

The data used to produce static maps consist of:

#### Vector data (shapefiles)

- Population in Greece (polygon): each administrative region (“OTA”) is connected to the corresponding population value (population data are referred to the year of 2001)
- Greek coastline (line)
- Neighbor countries (polygon)

### Software

Additionally, the whole process was done using QGIS software.

### Process

Starting the implementation process, our first task is the selection of an administrative region and at the same time, the creation of a new shapefile based on the data of the selected county. The above aims to create our data, which will be used to produce static maps.

Initially, the prefecture of Larissa was chosen to produce cartography products. From the population level of Greece, by opening the attribute table, we can process the data we need. The process included classifying the above table, based on its code (“OTA”), as well as creating a new shapefile with the data we need. In this way, we managed to create a new layer. Finally, a small processing of this product was carried out. The borders of the areas were changed in style and from a continuous line to a dash-dot-dot line. Below, the creation of this layer is shown.

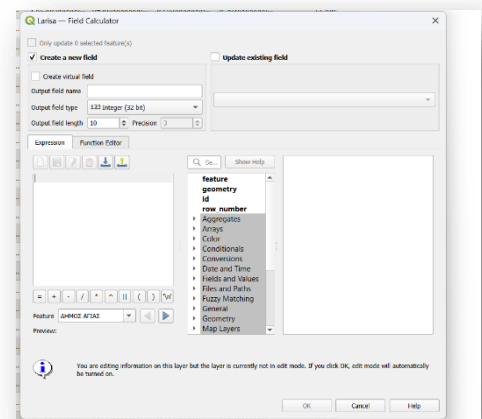


The next task was to calculate the population density. What had to be done was to intervene in the database, as this variable did not already exist. Having the total population and the total area of each area as data, the way to create the density is to divide the 2 characteristics above. Specifically, the mathematical formula is given:

$$Pop_{dens} = \frac{Pop}{Area}$$

This process was done with the help of the attribute table where I can edit my data using the "open field calculator" option. This option gives me the possibility to make changes even programmatically. Some screenshots of the process are provided.

| OBJECTID | CODE_OTA | NAME_OTA     | CODE_NOM | NAME_NOM    | AREA           | POP          | POP_AREA     | POP_DENS     |
|----------|----------|--------------|----------|-------------|----------------|--------------|--------------|--------------|
| 1        | 413      | ΔΗΜΟΣ ΑΡΓΕΕΣ | 42       | NOMOS ARGES | 122.2430000000 | 12679.000000 | 61219.000000 | 64857.000000 |
| 2        | 414      | ΔΗΜΟΣ ΑΡΓΕΕΣ | 42       | NOMOS ARGES | 196.5020000000 | 6458.000000  | 3228.000000  | 3238.000000  |
| 3        | 415      | ΔΗΜΟΣ ΑΡΓΕΕΣ | 42       | NOMOS ARGES | 154.1150000000 | 8407.000000  | 4204.000000  | 4723.000000  |
| 4        | 416      | ΔΗΜΟΣ ΑΡΓΕΕΣ | 42       | NOMOS ARGES | 143.8430000000 | 3951.000000  | 2087.000000  | 1864.000000  |
| 5        | 417      | ΔΗΜΟΣ ΑΡΓΕΕΣ | 42       | NOMOS ARGES | 102.9810000000 | 2273.000000  | 1125.000000  | 1148.000000  |
| 6        | 418      | ΔΗΜΟΣ ΑΡΓΕΕΣ | 42       | NOMOS ARGES | 51.3214000000  | 1022.000000  | 4876.000000  | 5046.000000  |
| 7        | 419      | ΔΗΜΟΣ ΑΡΓΕΕΣ | 42       | NOMOS ARGES | 113.0570000000 | 3119.000000  | 1570.000000  | 1548.000000  |
| 8        | 420      | ΔΗΜΟΣ ΑΡΓΕΕΣ | 42       | NOMOS ARGES | 292.9980000000 | 14641.000000 | 7185.000000  | 7378.000000  |
| 9        | 421      | ΔΗΜΟΣ ΑΡΓΕΕΣ | 42       | NOMOS ARGES | 161.1180000000 | 4526.000000  | 2314.000000  | 2212.000000  |
| 10       | 422      | ΔΗΜΟΣ ΑΡΓΕΕΣ | 42       | NOMOS ARGES | 96.2071000000  | 2428.000000  | 1217.000000  | 1211.000000  |
| 11       | 423      | ΔΗΜΟΣ ΑΡΓΕΕΣ | 42       | NOMOS ARGES | 128.6110000000 | 4175.000000  | 2201.000000  | 2174.000000  |
| 12       | 424      | ΔΗΜΟΣ ΑΡΓΕΕΣ | 42       | NOMOS ARGES | 143.5750000000 | 2834.000000  | 1432.000000  | 1402.000000  |
| 13       | 425      | ΔΗΜΟΣ ΑΡΓΕΕΣ | 42       | NOMOS ARGES | 162.8780000000 | 3305.000000  | 1642.000000  | 1601.000000  |
| 14       | 426      | ΔΗΜΟΣ ΑΡΓΕΕΣ | 42       | NOMOS ARGES | 205.1580000000 | 3174.000000  | 1681.000000  | 1581.000000  |
| 15       | 427      | ΔΗΜΟΣ ΑΡΓΕΕΣ | 42       | NOMOS ARGES | 179.6400000000 | 1763.000000  | 884.000000   | 878.000000   |
| 16       | 428      | ΔΗΜΟΣ ΑΡΓΕΕΣ | 42       | NOMOS ARGES | 160.0100000000 | 3187.000000  | 1615.000000  | 1572.000000  |
| 17       | 429      | ΔΗΜΟΣ ΑΡΓΕΕΣ | 42       | NOMOS ARGES | 107.8760000000 | 2976.000000  | 1532.000000  | 1444.000000  |
| 18       | 430      | ΔΗΜΟΣ ΑΡΓΕΕΣ | 42       | NOMOS ARGES | 197.5300000000 | 3472.000000  | 1778.000000  | 1694.000000  |
| 19       | 431      | ΔΗΜΟΣ ΑΡΓΕΕΣ | 42       | NOMOS ARGES | 165.7010000000 | 1781.000000  | 952.000000   | 829.000000   |
| 20       | 432      | ΔΗΜΟΣ ΑΡΓΕΕΣ | 42       | NOMOS ARGES | 172.4380000000 | 5486.000000  | 2892.000000  | 2684.000000  |
| 21       | 433      | ΔΗΜΟΣ ΑΡΓΕΕΣ | 42       | NOMOS ARGES | 278.1560000000 | 6540.000000  | 3271.000000  | 3269.000000  |
| 22       | 434      | ΔΗΜΟΣ ΑΡΓΕΕΣ | 42       | NOMOS ARGES | 304.9650000000 | 2588.000000  | 1811.000000  | 1775.000000  |
| 23       | 435      | ΔΗΜΟΣ ΑΡΓΕΕΣ | 42       | NOMOS ARGES | 245.4620000000 | 8292.000000  | 4132.000000  | 4160.000000  |
| 24       | 436      | ΔΗΜΟΣ ΑΡΓΕΕΣ | 42       | NOMOS ARGES | 292.4080000000 | 6412.000000  | 3424.000000  | 2988.000000  |



In the third part of the paper, we are asked to characterize the measurement scale, for each characteristic of the given population model. We know from theory that measurement scales are divided into the following 4 categories:

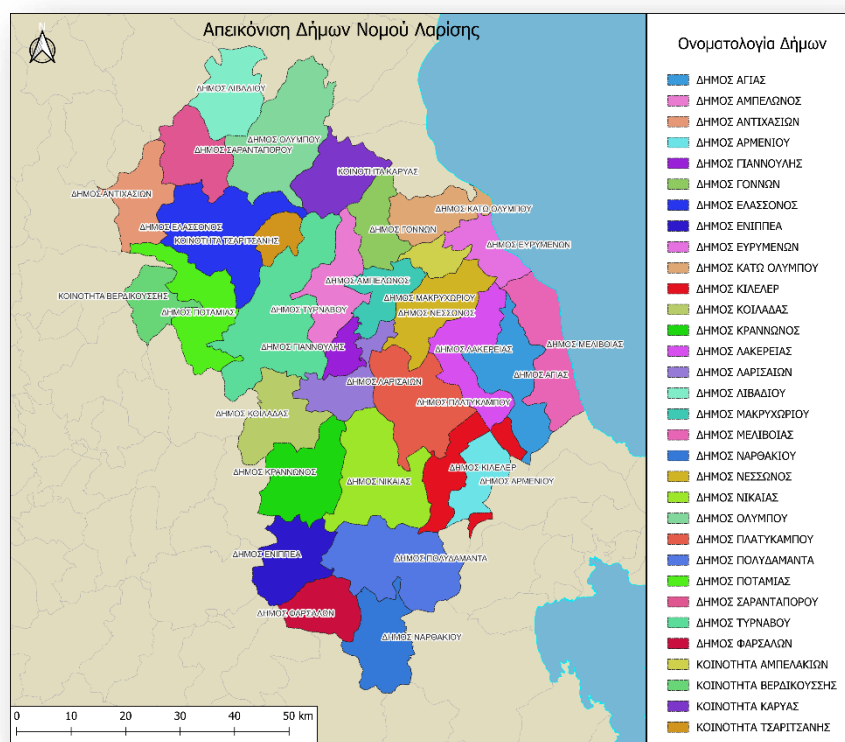
1. Nominal: Where we have categories without order or numerical significance.
2. Ordinal: Categories in order or rank, but with no known differences between the data.
3. Interval: Measurements with fixed intervals, but no real zero and
4. Ratio: Measurements with numerical differences and absolute zero.

Looking at the table shown above, we can see that its characteristics are:

1. OBJECTID: Ordinal  
It represents the order of regions, and there is a concept of ordering.
2. CODE\_OTA: Nominal  
This code represents a specific region, used for categorization without numeric relationships.
3. NAME\_OTA: Nominal  
This is a text-based attribute representing the names of regions.
4. CODE\_NOM: Nominal  
Like CODE\_OTA, this code categorizes a specific attribute (nomos) without a numerical relationship.

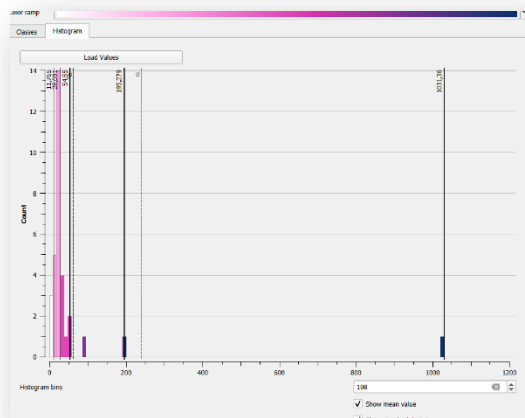
5. NAME\_NOM: Nominal  
This is a text attribute that represents the name of the prefecture, used for categorization.
6. AREAk2: Ratio  
The area is measured in  $km^2$  and there is absolute zero, i.e. no area.
7. POP: Ratio  
And here there is the real zero, i.e. the absence of population.
8. POP\_FEMALE: Ratio  
Like the previous one, but it only applies to the female population.
9. POP\_MALE: Ratio  
And in this case, it applies to the male population.
10. POP\_DENS: Ratio  
Here we have measurements with numerical differences, but also absolute zero which means no population, as without population we cannot have density.

In the next step, we are asked to visualize the administrative regions ("OTA") within the Prefecture of Larissa. The data used are those of the first step, i.e. the new layer created for the data of Larissa Prefecture. In this case, however, we categorized the areas based on NAME\_OTA. The visual variable of color hue for area was used. Based on the guides we had in theory; we know that the data belonging to Nominal scale is suitable to be used in the specific variable chosen. With this process, we achieved the correct visualization of our data, but also the categorization method helped us to choose the appropriate labels. The colors chosen were random. By applying the above procedure, we were able to produce the map shown below:

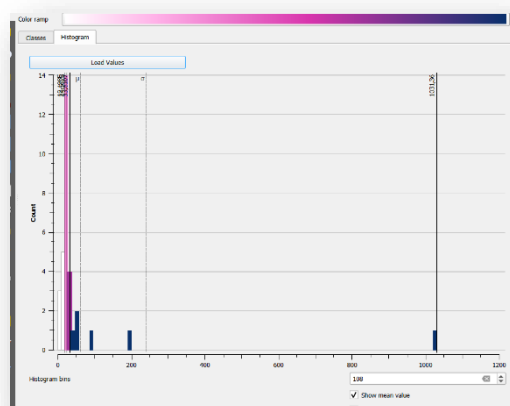


The next map to be produced is the visualization of the population density distribution. The map to be generated must be choropleth. In other words, a choropleth map is used to visualize quantitative data, but of derived variables. The data we need is the data generated in step 2, namely the population density. In this case too, a categorization of the data had to be done, but this time the population density was used as a value. Also, because clustering had to be done, we had to use the visual variable of color value.

In this step, the method chosen was Natural Breaks (Jenks), which is ideal when the data have large differences and 5 classes were used. The QGIS software provides us with several different categorization methods. Some of them are Equal Interval, Quantile, Standard Deviation etc. Usually, the first method is used when the values are relatively evenly distributed, which in our case is not the case, as when observing the histogram of the values we notice large variations. The second method was the second most appropriate, as it is used for isometric categories in terms of areas. The last method is used to highlight deviations from the meaning, which we did not need in this case.

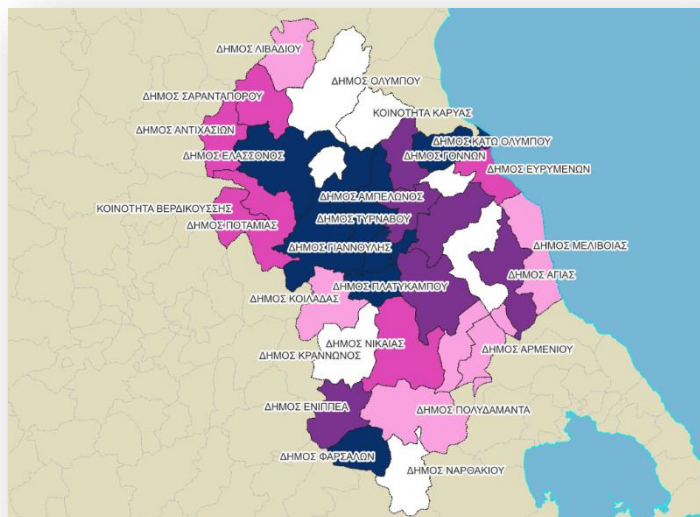
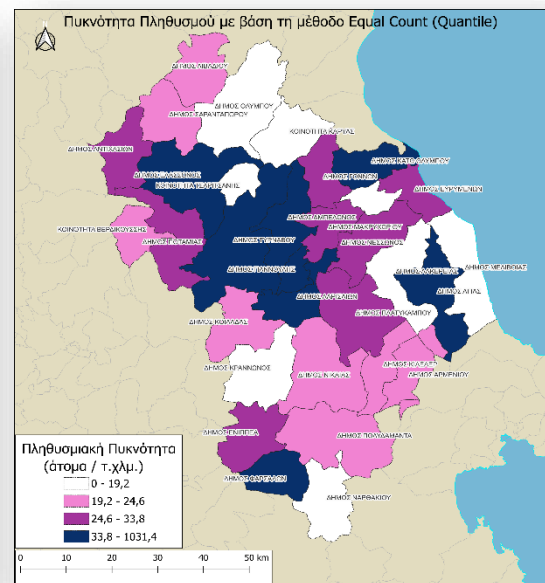
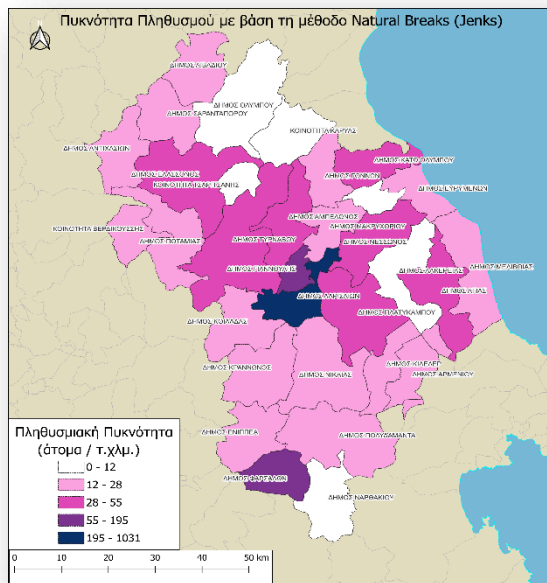


So, in our case, some observations were made. Based on the table, it was observed that the population density has large variations. This indicates to us a relatively asymmetric distribution and the presence of many outliers. That is, we have an accumulation of small values, with a few areas having a high density, such as urban centers. In the histogram image, we can see exactly what we saw in the data table.



On the other hand, there was also the option of choosing the Quantile method. While this method offers an equal distribution of areas in the categories, it was observed that areas with very different population density values were placed in the same category, creating a less realistic map. Thus, the result was less representative of the actual distribution, especially for areas with extreme values.

As a result, the map generated by the Natural Breaks method presented a more reasonable and realistic distribution of areas, especially for identifying high density areas, such as urban areas compared to low density areas, i.e. rural areas. In contrast, the map produced by the Quantile method was more balanced in terms of the number of areas per category, but less indicative of the distribution of population density. The corresponding maps are attached.

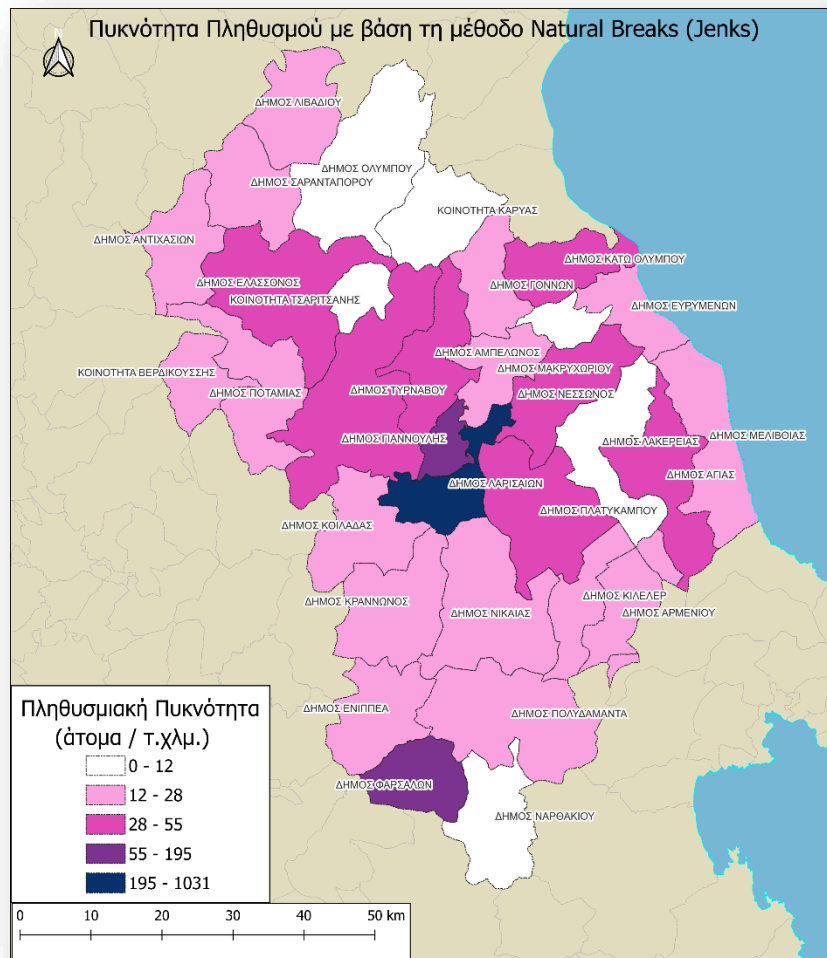


Also, a final attempt was made to do a categorization with 5 classes in the Quantile method, but the result did not change much.

Finally, the way the classes were selected in the final map was purely by observation, as with 5 classes there was a very good mapping of values.

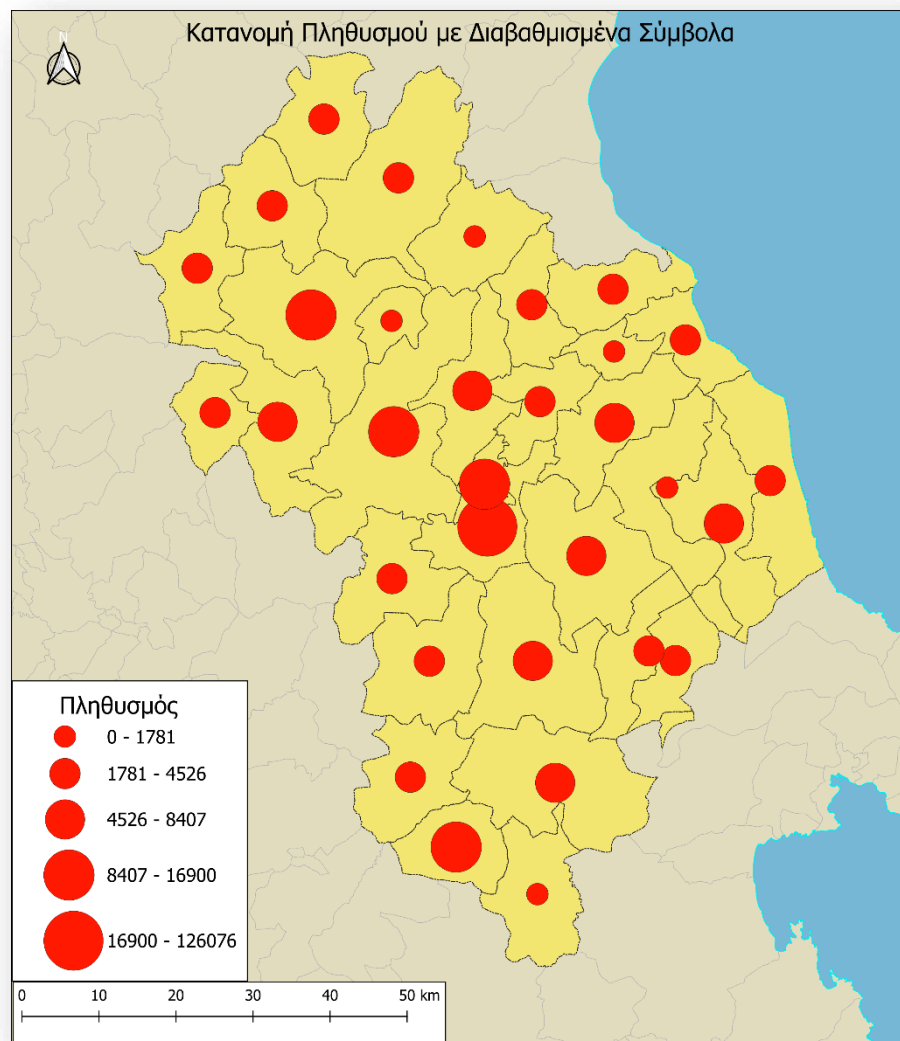


Here is the final map.

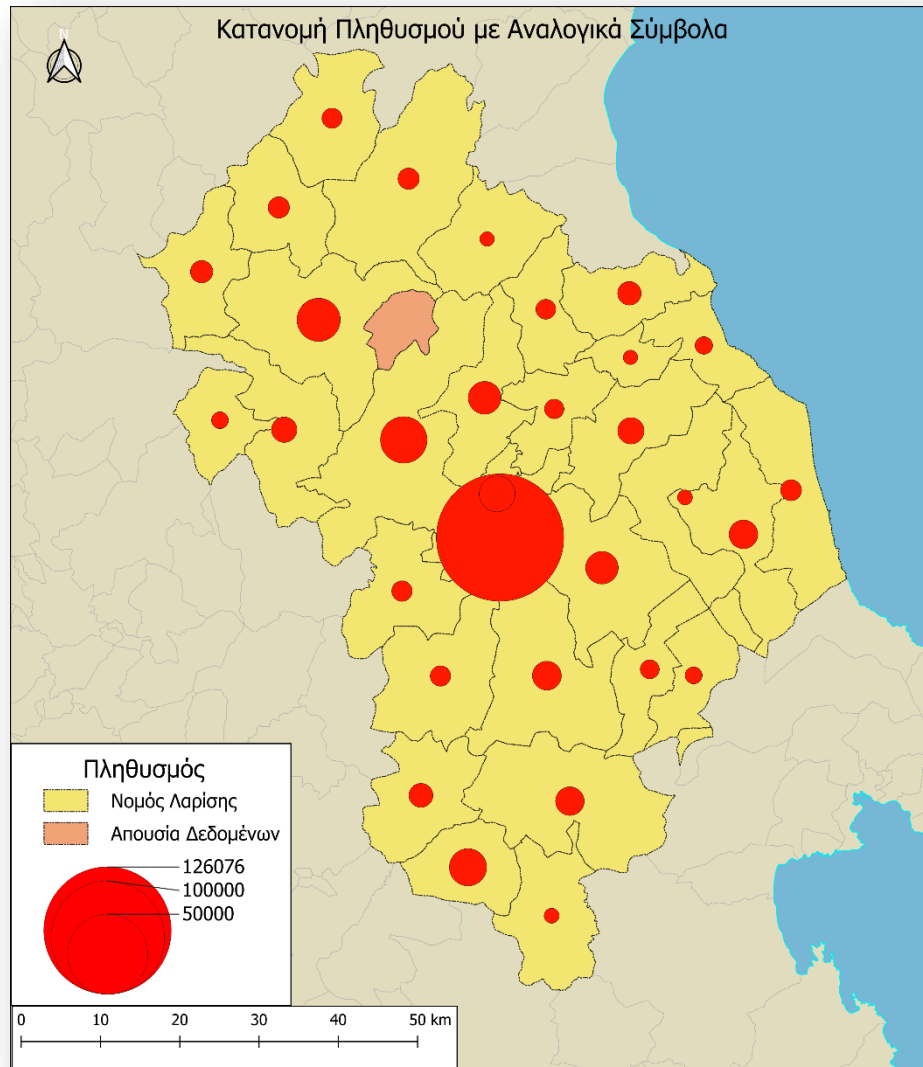
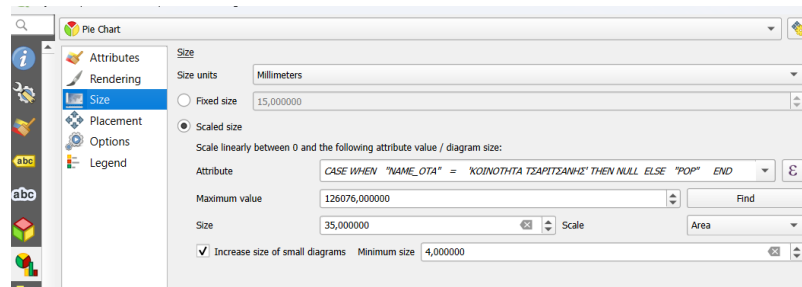


The next map we were asked to produce is based on the visualization of the population distribution with graduated symbols. Essentially, a similar process to the previous map was used in the production of this map. As data, we had those relating to the total population for each region. In this case, we used the visual variable of size. This variable is suitable for visualizing quantitative differences. A size classification was made, where an area value was assigned to each class. The application as well as the classes used are the same as before, i.e. 5 classes and the Natural Breaks (Jenks) method. This choice was made in order to have the correct wording between the two maps.

Here is the third map.

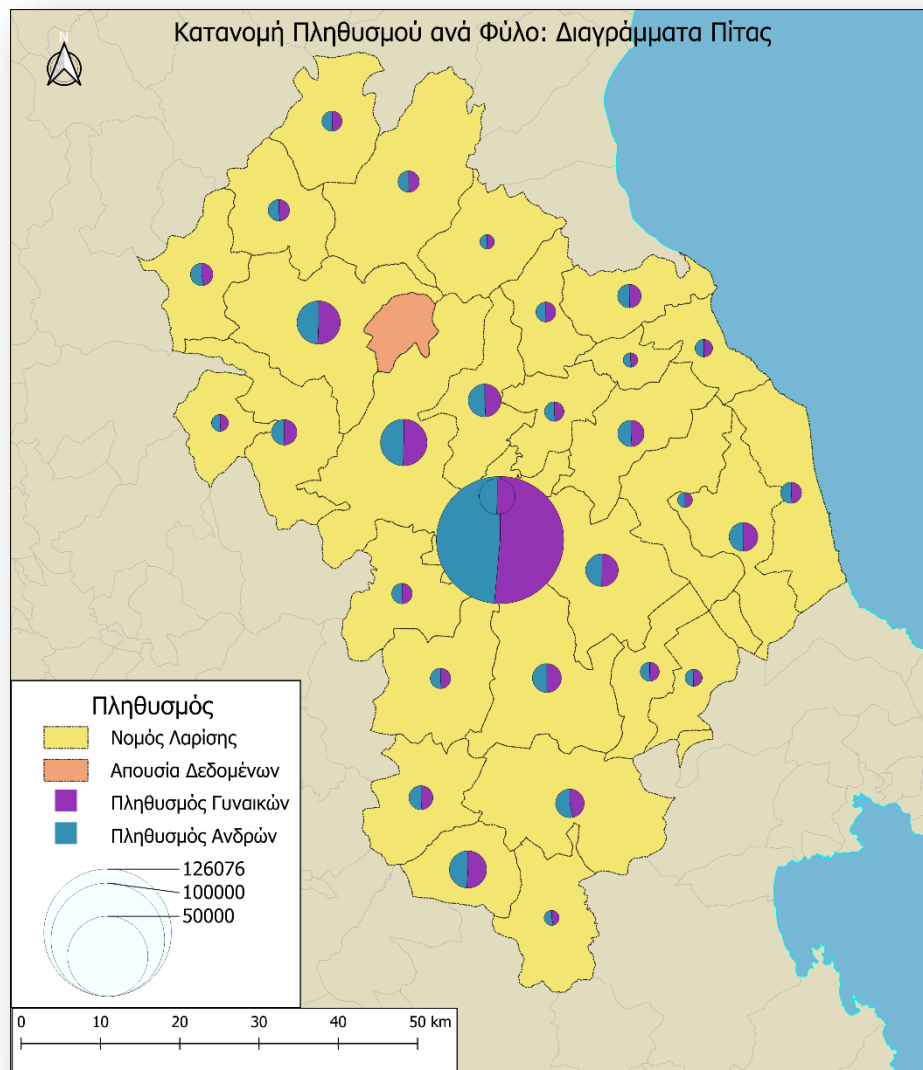


Then we had to create the same map, but this time with proportional symbols. In this case too, we have a visualization of quantitative data, as our data are the population values. The visual variable of size was used. Our data, is primary and this gives us the opportunity to use proportional symbols. Essentially, there is an increase or decrease in area depending on the different values. It was applied by creating pie charts, with the size of the population value. Finally, in processing the size of the symbols, a modification was made as one area was not supported by data. In this way, we removed this region from the symbol construction. As a result, the resulting cartographic product is given below. The command used for the different visualization of the area without data is also given.

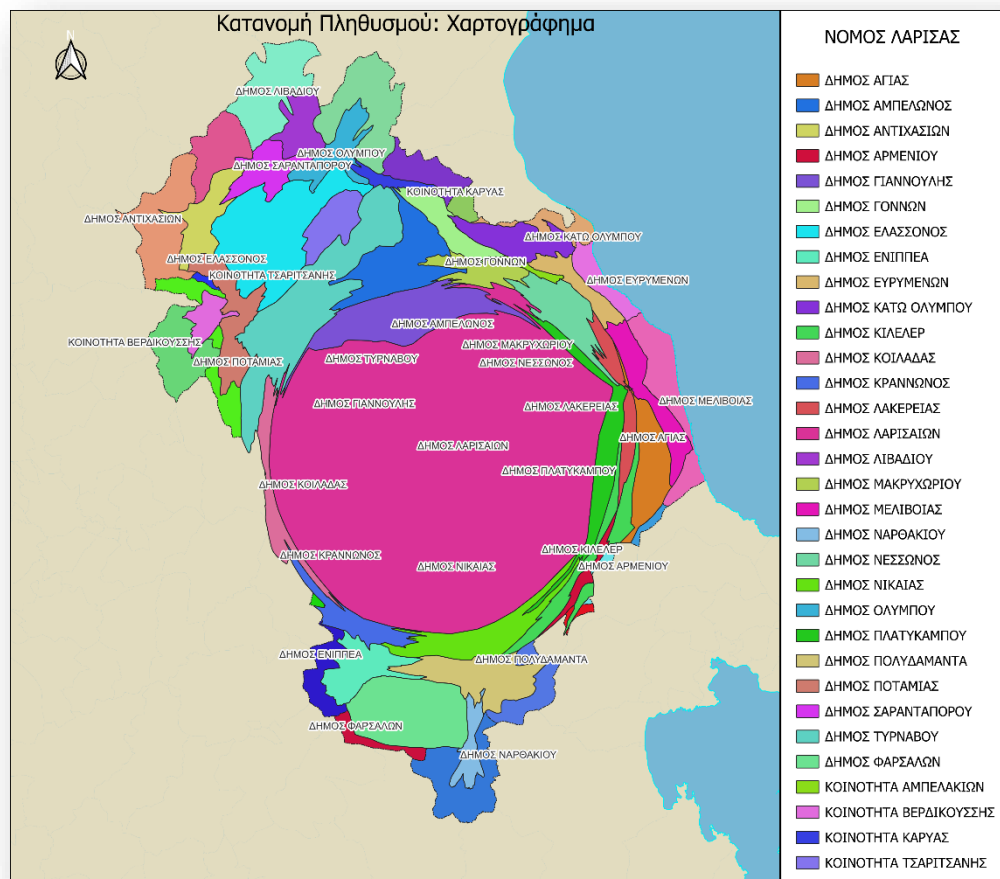


The next map is a visualization of the population distribution considering gender information using pie charts. The same procedure was followed as for the previous map, except that the pie charts were constructed based on the population sizes of women and men. The variables and the application remained the same. The map created was as follows.





The last map to be created was the visualization of the population distribution using a cartogram. Essentially, a cartogram is a special type of map that distorts geographical areas according to the size we want to show. In this case, the size is the population. Specifically, a Contiguous Cartogram was created. To create the cartogram, we had to add a new plugin cartogram3 to the QGIS software. Additionally, in creating this map the boundaries no longer made sense, so we created a duplicate layer that was about population, but this time without boundaries. Finally, a classification was done as far as the visualization of the different areas is concerned, exactly in the same way as the production of the 1st map. Therefore, the map created is as follows.



The last part of the paper is the discussion of different methods that can be used to visualize the population distribution and the population density distribution, knowing that both are quantitative data. A Dasymetric map could be created, instead of a Choropleth map, which uses intensities and starts from the boundaries of the class where the data is generated, but it is about secondary data. Also, a 3D map could be created which uses heights to show the distribution of either population or density.

Another map suitable for primary data is the Dot map. In this case, 1 dot corresponds to 1 value of the phenomenon and is randomized so we cannot get exact values. Still, a Heat map would be appropriate as it uses a gradient of colors to indicate intensity, with warmer colors representing higher population or density levels. Also, maps that use the visual variable of size by changing the thickness of the lines, as well as maps based on the visual variable of texture, would be suitable for visualizing the density distribution. Finally, the Cartogram can also be used for density but in different formats. These are Non-Contiguous and indeed either Overlapping, or Non-Overlapping. Finally, the Dorling Cartogram could be used, where the area surfaces are replaced with geometric symbols.

# Cartographic Visualization

## 2<sup>nd</sup> Exercise

Subject: “Production of an Interactive Web Platform”

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### Description

This work consists of the production of an interactive web platform which included the above components:

- An animated map: Visualization of Peloponnese’s population for the period 1940-2011.
- An interactive surface (3D) map: Visualization of both elevation and depth information in Santorini Island and
- A multimedia interactive map: Visualization of airport locations in Greece and connection to multimedia material (e.g. web pages etc.)

### Data

The data used to produce an interactive web platform is as follows:

Vector data:

- Polygons of administrative regions (“NOMOS”) in Peloponnese connected with the correspond population data.
- Centroids of administrative regions (“NOMOS”) in Peloponnese connected with the correspond population data.
- Peloponnese Coastline.
- Polygons of neighbor regions.
- Study area outline.
- Airport’s locations in Greece.

Raster data:

- Digital Elevation Model (DEM) of Santorini Island.
- Digital Elevation Model (DEM) around Santorini Island.

### Software

In addition, the process was carried out using QGIS, GIMP and Kompozer software. Also, two new plugins were added to QGIS, Qgis2threejs and Qgis2web.

### Process

The process starts with the production of the Animated map, where the population of the Peloponnese is visualized for the period 1940-2011. The data used are all the

Vector Data mentioned above, except the last one which refers to the airports of Greece. First, I had to use QGIS to create, for each period, the population distribution map for the Peloponnese. A small edit was made to my original data, in terms of colors and boundaries between regions.

Then, to produce the population distribution maps for each chronology, centroids had to be created. This implies that a categorization with graduated symbols has to be done and this is because our data is quantitative data. The same process could be done with proportional, but I preferred the graduated ones as I believe they maintain greater visual clarity.

The problem is that the final map we want to produce is Animated. This requires changing only the symbols, as well as the display date that will be on the map generated each period. Therefore, to create the appropriate categories, we cannot take the data for each year separately, but all the data must be properly processed. That is, the categories created should be based on the minimum value, of all minimum values, as well as the maximum value, of all maximum values.

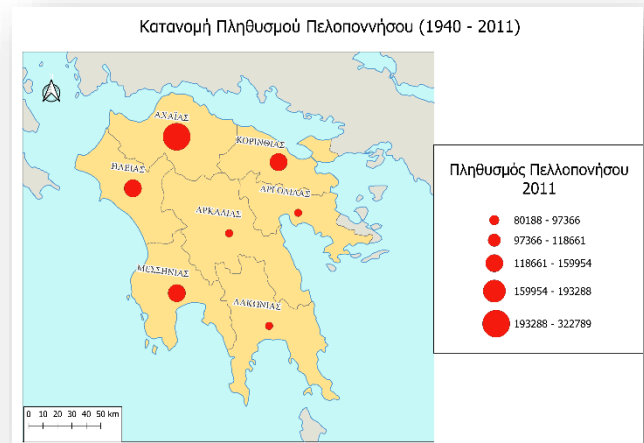
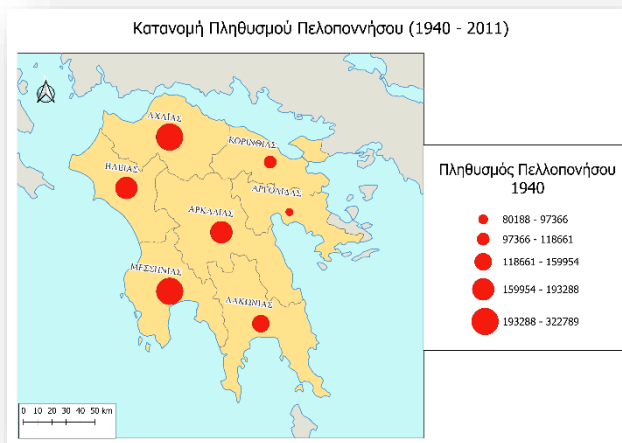
To solve this problem, we created a code in Python programming language, where with the help of the function "quantile", we managed to find the values of the data, corresponding to specific percentages of their distribution. Below is the code created in the form of a screenshot.

```

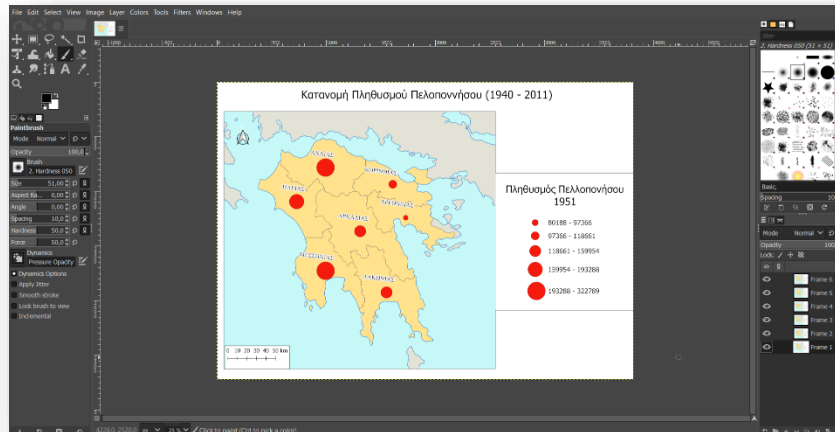
1  @author: Bantisouli Agathi
2  """
3  #Library
4  import numpy as np
5
6  #Import the dataset as a NumPy array.
7  data = np.array([
8      [308694, 388078, 275164, 239859, 239286, 228871, 159954, 166964, 159953, 173877, 211970, 227871, 240355, 176876],
9      [159300, 179429, 160257, 165956, 188861, 188274, 185085, 193288],
10     [145082, 141823, 123085, 113115, 112505, 113358, 111547, 154624],
11     [97044, 97366, 92742, 88698, 90145, 85389, 80188, 105770],
12     [89138, 95696, 93188, 95844, 118661, 130898, 144156, 99637],
13     [86685, 105309, 108096, 111263, 135042, 154361, 171062, 102035]
14 ])
15
16 #Calculate the quantile limits for the data.
17 quantile_limits_5 = np.quantile(data, [0, 0.2, 0.4, 0.6, 0.8, 1])
18
19 #Print the calculated quantile limits.
20 print("Quantile limits:", quantile_limits_5)

```

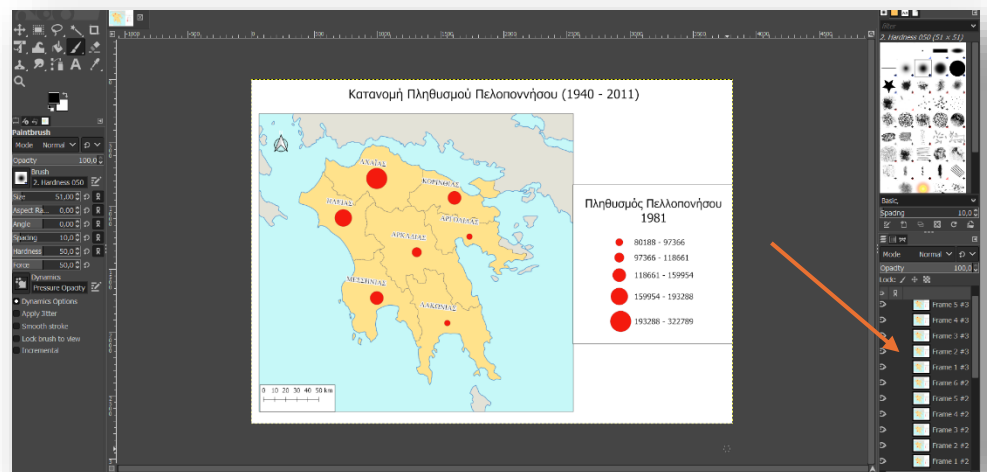
Therefore, after testing, the classes chosen were 5 in number, as they best represent the data values, since there is a lot of variation. Finally, the method used was Quantile, as we also wanted to have a balanced comparison between the regions, something that the other methods do not provide. In the selection process, the 1st paper also helped, as the process of producing the maps was the same. Finally, the labels were added to show the regions of the Peloponnese. Here are 2 maps produced.



After all these maps were produced, they had to be compiled. Therefore, given all the maps and with the help of the GIMP software, to achieve this process I used the blending function. Specifically, I imported two images in separate layers and created a copy of the first one. I then applied blending techniques to create a smooth transition between them. This process was repeated for each pair of images, creating a unified and cohesive effect. An example of blending is given for the first 2 maps.

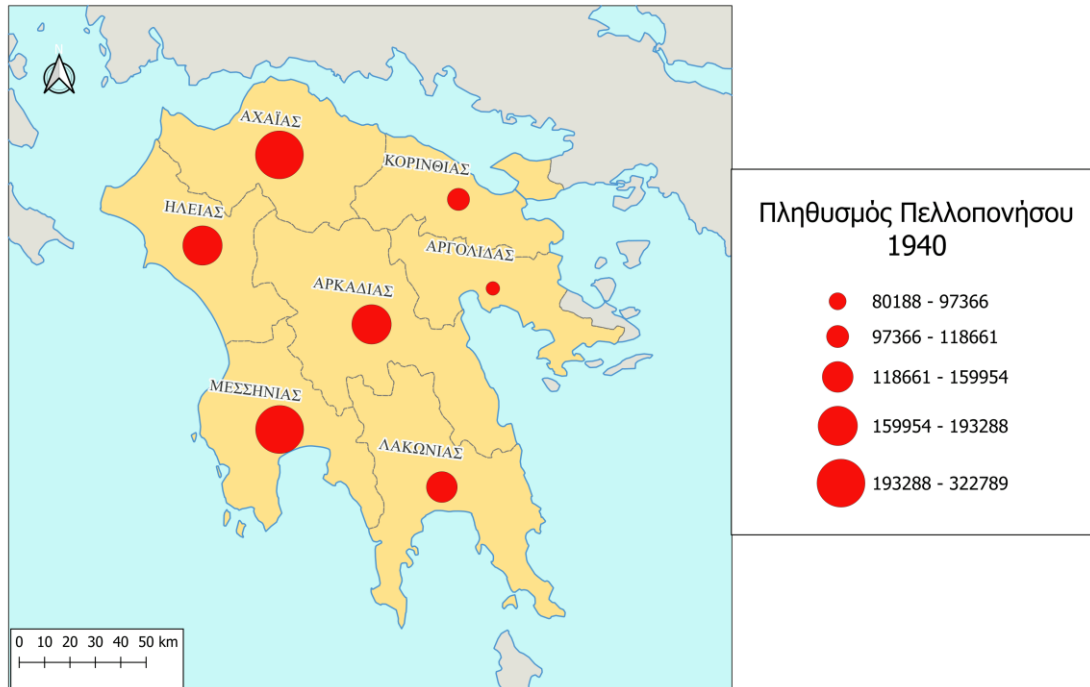


Then I opened all the generated blending files as layers. So, I had one file with all the frames created. A small example is given in the opposite image.



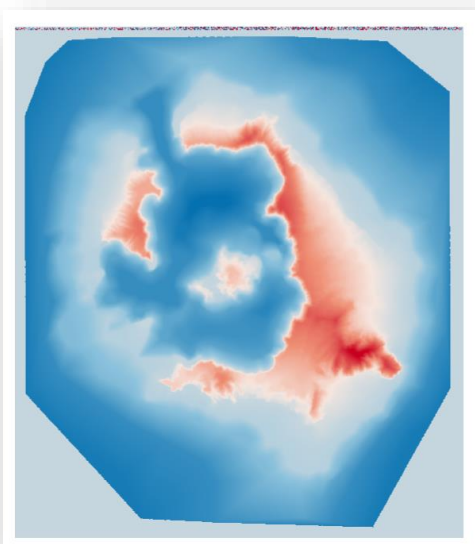
Finally, what had to be done is to export the result as a .gif file, while making some adjustments, i.e. to constantly repeat the process. And the result is given below, but in a static product.

### Κατανομή Πληθυσμού Πελοποννήσου (1940 - 2011)

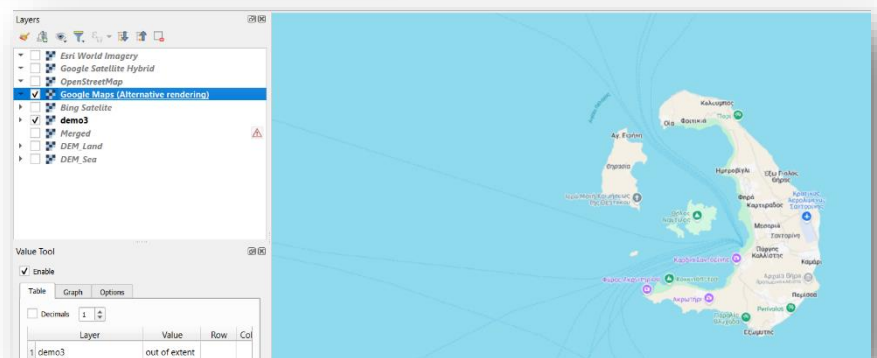
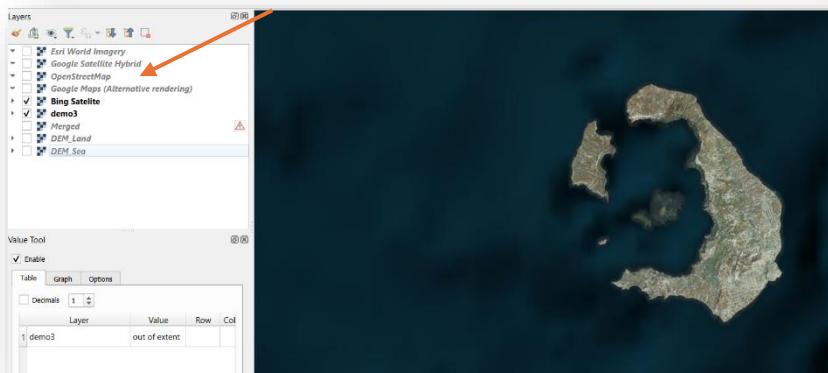


Then the second map that had to be produced was the Interactive map (3D). As data, we had the rasters mentioned at the beginning. One refers to Santorini and the other to the sea around the island. Basically, the raster models provide us with data for each point. Specifically, using QGIS and importing this data, we can see for each point on Santorini, the elevation of the base of the particular model, and also the depth of the sea.

In the next phase, what we did was to merge our two models into one. This was done with the help of QGIS software. This created a single model, which gives us both altitudes and depths. In addition, with the help of the "QuickMapServices" plugin several services were added to the single model. My choices were "Esri World Map", "Bing Satellite", "Google Maps", "Google Satellite Hybrid" and "OpenStreetMap". We can also see some examples.







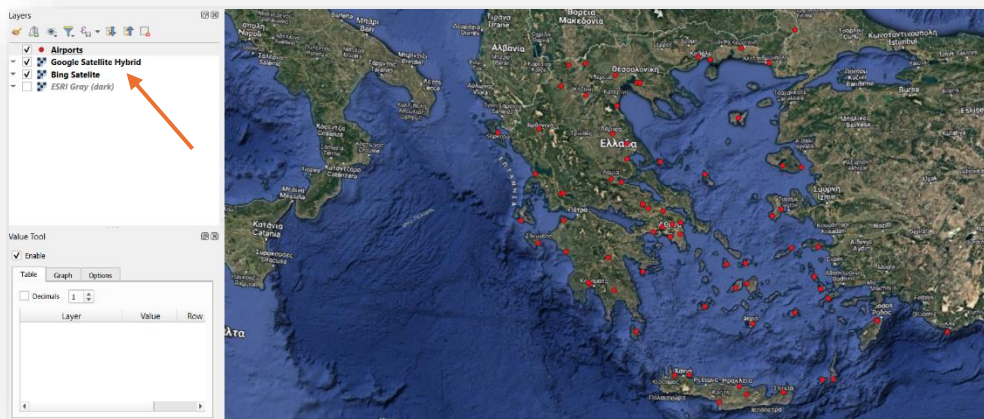
The next step was the construction of the 3D map. With the help of the plugin "Qgis2threejs" and after some modifications, such as the scale of the map and its boundaries. 2 such maps were created. One is a template for mobile devices, while the other one is general. In the second map, an animation was also created, where the user when opening the file, will be able to see it.



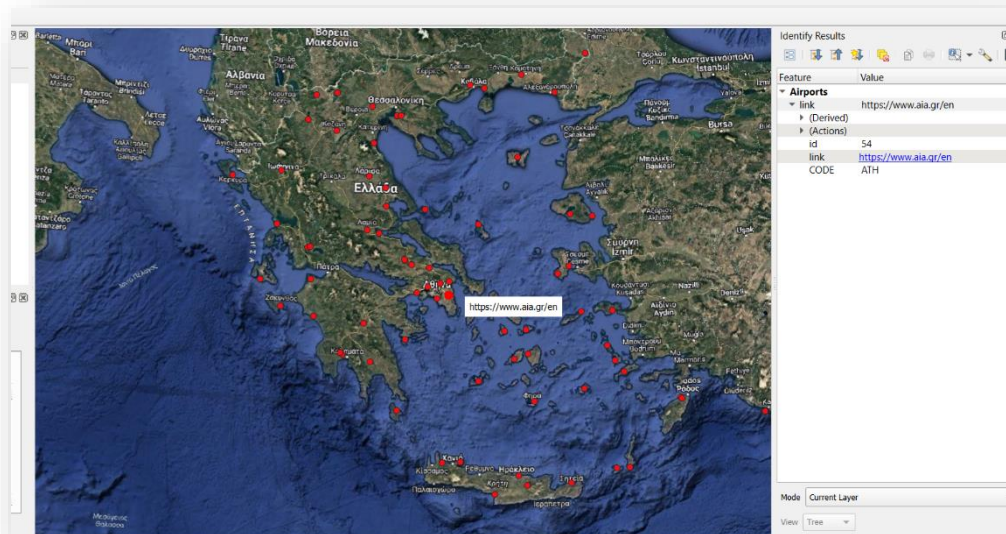
We can also see the final product in a static map, because normally our files are in .html format.



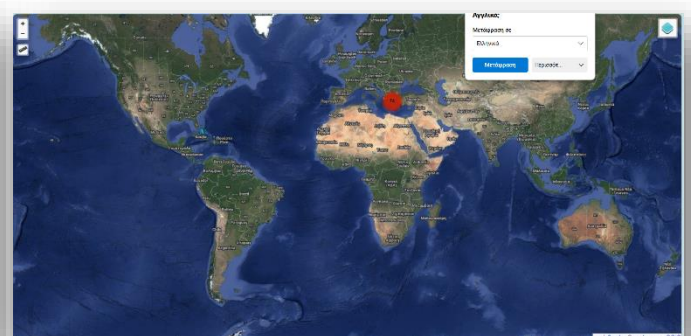
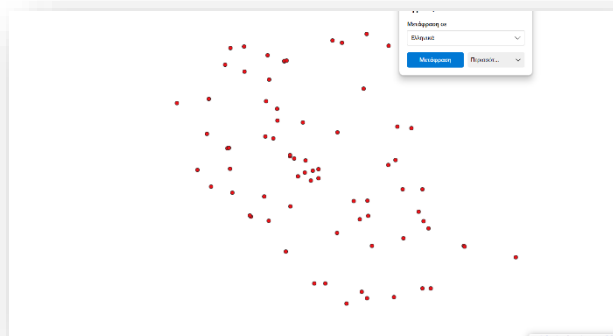
The last map to be made was a multimedia interactive map. It had to be visualization of airport locations in Greece and connection to multimedia material (e.g. web pages etc.). Having the airport locations as data, the map was created with the help of



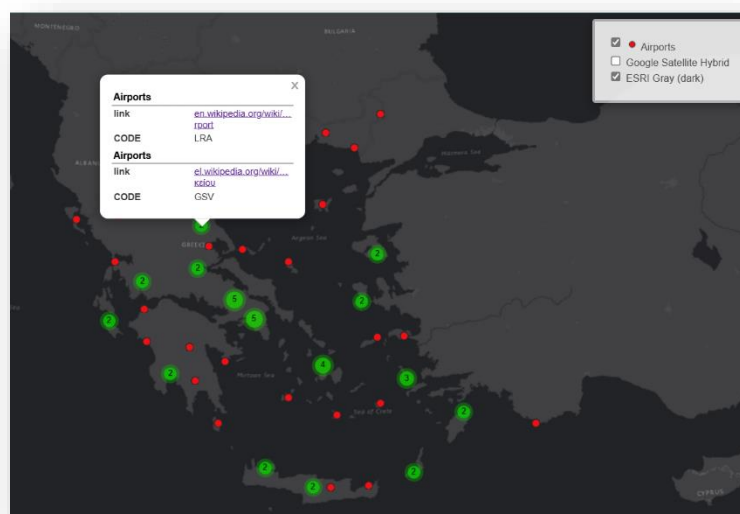
QGIS. As before, I introduced several services in my work. Then what I did was to edit each point and add the necessary information that was needed. That is, I added the websites, as well as their abbreviations to IATA (International Air Transport Association) or ICAO (International Civil Aviation Organization) if they were state airports. Some data was left blank as I was unable to collect information or due to military airports it was not possible.



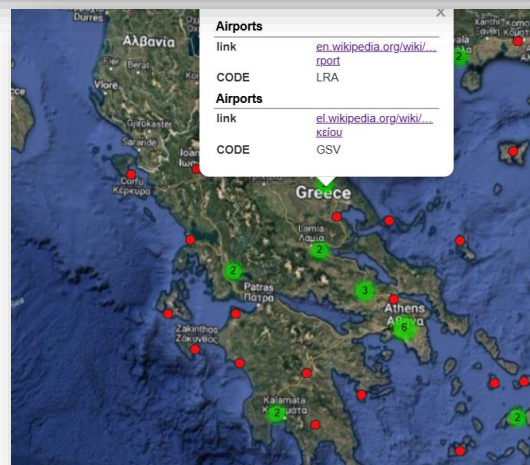
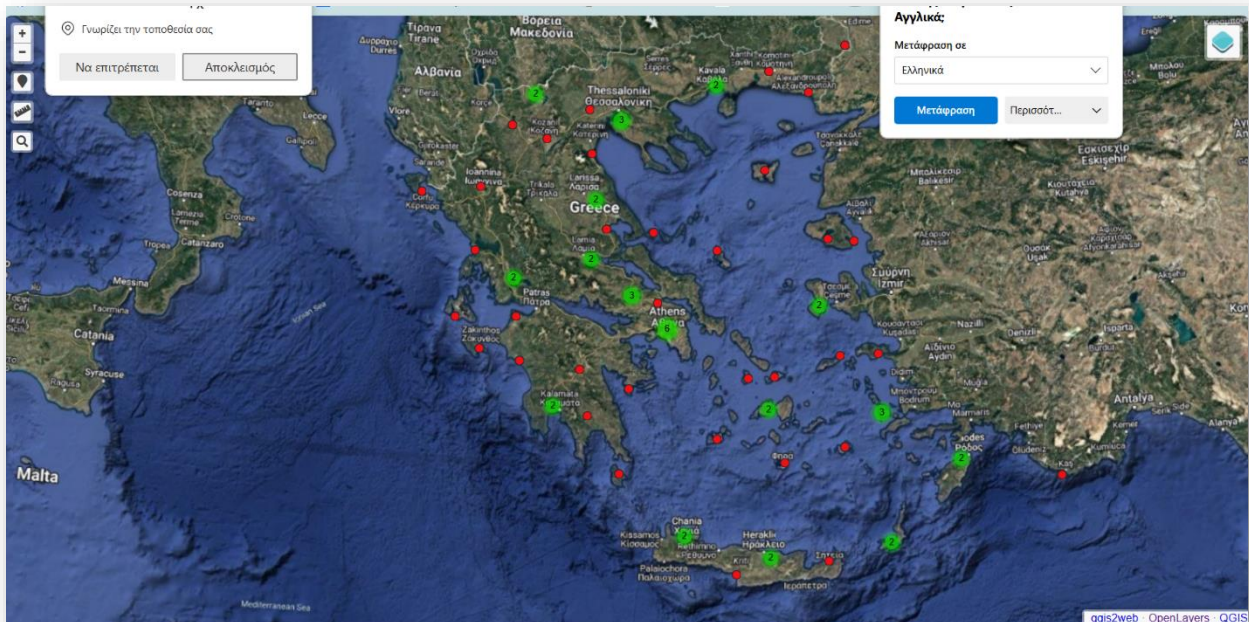
Finally, the web map had to be created. With the plugin "Qgis2web" and after many attempts and tests, I managed to create the desired map. Here are indicatively two attempts, and the final result.



As we can see in the result, a grouping was also done so that the reader can see at a glance the number of airports located in a region. Also, it is possible to locate an area with the help of search button, but also various other actions.

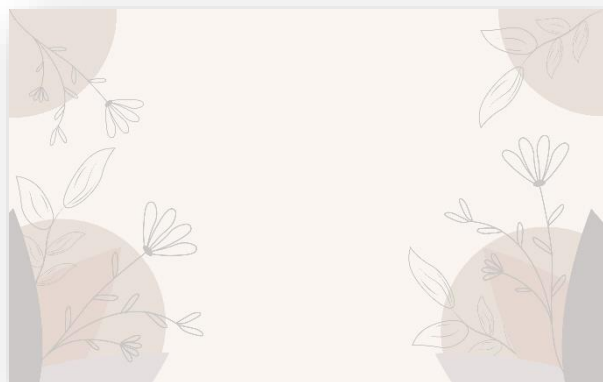




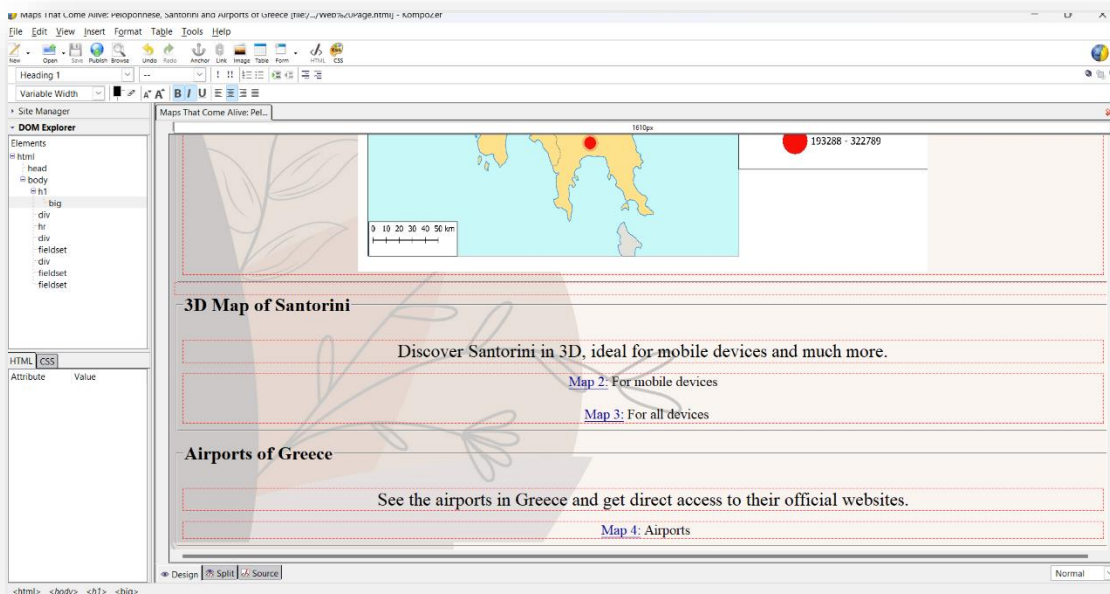


The last part of the project was the construction of a website, with the 4 maps created in the previous steps. The construction was done in Kompozer software, but also in GIMP, where the background was edited.





With the help of Kompozer, a complete product has been created, where all 4 maps are provided, with the last 3 being on the page in the form of a link.



The final product is in .html format and therefore some screenshots are given.

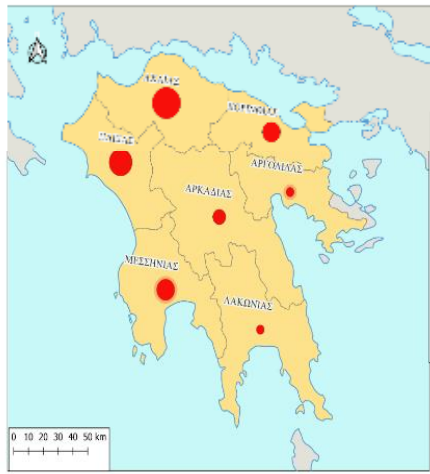
### "Maps that come to life: Peloponnese, Santorini and Airports of Greece"

Discover Greece through three unique interactive maps: Track the population distribution of the Peloponnese (1940-2011), explore Santorini in 3D and find all the airports in Greece with access to their websites.

#### Population Distribution of Peloponnese (1940 - 2011)

Follow the population distribution of the Peloponnese with an interactive visualization.  
Map 1:

Κατανομή Πληθυσμού Πελοποννήσου (1940 - 2011)



| Πληθυσμός Πελοποννήσου 1991 |
|-----------------------------|
| 80188 - 97366               |
| 97366 - 118661              |
| 118661 - 159954             |
| 159954 - 193288             |
| 193288 - 322789             |

#### 3D Map of Santorini

Discover Santorini in 3D, ideal for mobile devices and much more.

[Map 2:](#) For mobile devices

[Map 3:](#) For all devices

#### Airports of Greece

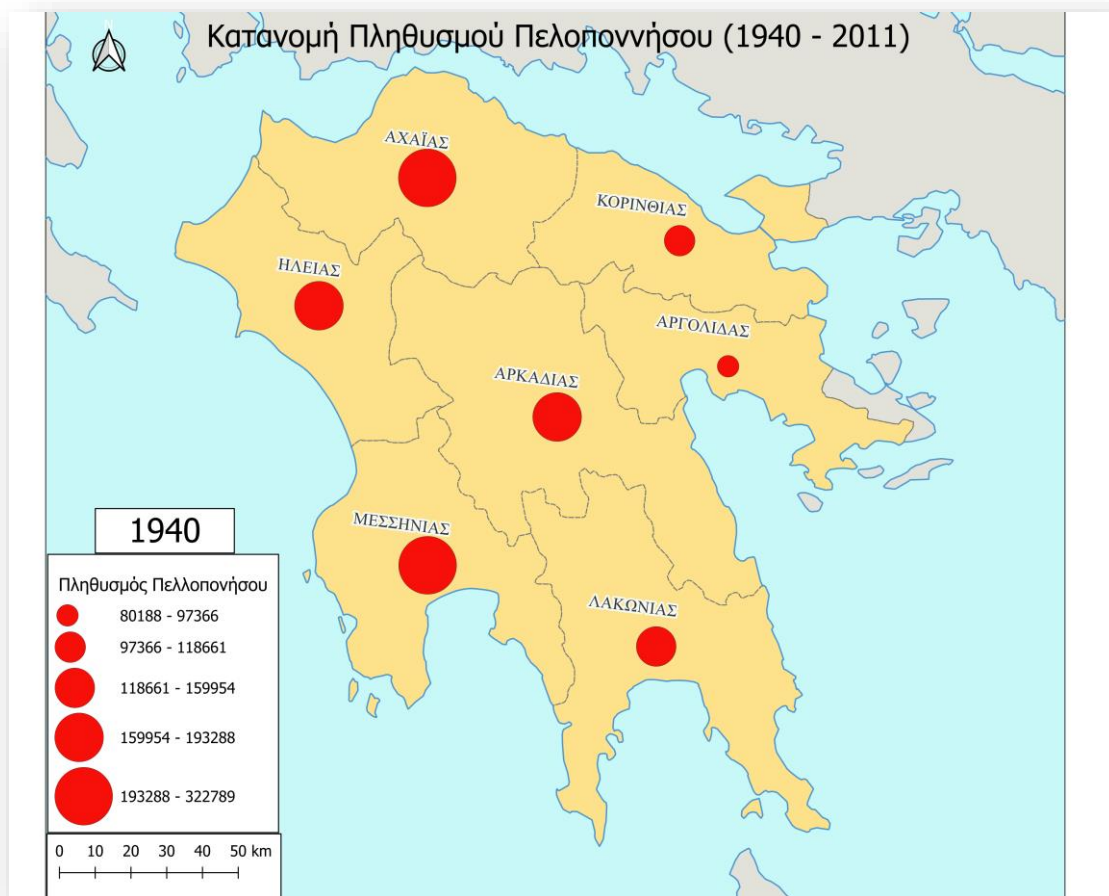
See the airports in Greece and get direct access to their official websites.

[Map 4:](#) Airports



Supplement: After observation, a change had to be made to the 1st map. The position of the caption was wrong, so the appropriate modifications had to be made. The process was done again from the beginning, up to the construction of the final .gif file.

Below, the final file is presented in static format, and the final product is also given.



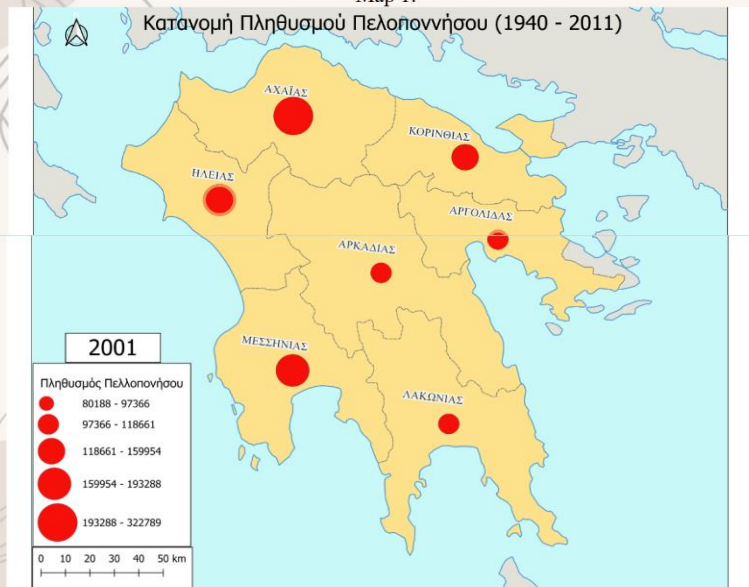
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### Airports of Greece

See the airports in Greece and get direct access to their official websites.

[Map 4:](#) Airports

# Cartographic Visualization

## 3<sup>rd</sup> Exercise

Subject: "Semester Exercise"

Bantisouli Agathi AM: aivc24010

### Description

In this task, an object must be selected. My choice involves the analysis of a research article and literature review of the related state of the art. The selected article is: "Eye-tracking Evaluation of Weather Web Maps" by Popelka et al.

### 1. Introduction

Nowadays, web maps play a significant role in day-to-day life. Information related to spatial and weather conditions can be obtained by accessing these. They have resulted in helping them a lot in their normal life. It is normal for the people to get instant information about weather conditions or any traffic jams or road congestion. In this way, web maps help an ordinary user to plan his journey or to wear clothes according to the weather forecast.

On the other hand, the usefulness of web maps stretched down to the professional sphere, including experts in geoinformatics and cartographers. Professionals rely on these resources in data analysis and decision making. Constant development efforts raise web maps to the next level. New features such as dynamic visualization, predictive analytics, or interactive layers add to their value.

With these developments, however, come other kinds of challenges. Increased complexity in map interfaces translates to reduced usability, particularly for unsophisticated users. Experts have to put up with unclear labels, too much visual information and unfavorable layout which impairs effective orientation. It is about time these usability challenges are dealt with. This way, web maps can serve the purpose of a wide array of users effectively.

It is of great importance to understand how people use web maps. Eye-tracking and user interaction studies reveal how users explore and understand the content of maps. Such studies can highlight usability issues, such as difficult to understand labels or comprehensiveness of visual elements. The research also points out shortcomings in the layout of maps and interactions with dynamic features. Resolving these issues will better position web maps to meet the needs of both the general user and the specialist.

The paper presents an analysis of the article "Eye-tracking Evaluation of Weather Web Maps" by Popelka et al. Further, it includes a review of the current research

related to the topic. This aims at underlining the contribution brought by the article to the study of usability within cartographic applications. Furthermore, this analysis aims to outline the lacunas of literature. The suggestions for future research will be targeted at the enhancement of web map design. Ultimately, this analysis stresses the need to understand how users interpret and interact with maps. This understanding is crucial in the development of effective cartographic products.

## 2. Analysis of the Selected Article

### 2.1 Purpose and Objectives

The research carried out by Popelka, and colleagues focuses on exploring the usability issues found in web-based weather maps. This study employs advanced methods, including eye-tracking technology, to observe how users engage with these maps. Additionally, it incorporates the think-aloud protocol, where participants express their thoughts while interacting with the maps. The main goal of the study is to gain a deeper understanding of how users interact with selected weather maps.

Another important aim is to identify specific usability problems that may prevent users from accurately interpreting the information presented on these maps. These issues could range from unclear visual elements to confusing layouts, both of which can hinder users from obtaining critical weather information. By pinpointing these challenges, the research seeks to offer clear recommendations for improving the usability of web weather maps.

Enhancing usability will help ensure that users can easily access and understand weather data. The findings from this study could lead to better-designed maps that provide clear and useful information, ultimately benefiting users seeking reliable weather forecasts.

The objectives of the study are multifaceted, targeting both practical usability concerns and theoretical contributions to the field of cartography.

- **Understanding User Behavior:** The study investigates how different categories of users, such as novices and experts, approach and utilize weather web maps. Through this research, an analysis of how these maps were constructed and the identification of their strengths and weaknesses, the decision-making processes, i.e. the strategies followed by the users and the ability to effectively identify key information.
- **Identifying Usability Difficulties:** The main objective of this work is to identify the difficulties that users, whether experts or non-experts, face when interacting with weather maps. The main difficulties identified related to poor quality captions, misplacement of main tools that help to understand the maps, complex cartographic

products in terms of their design, and poor interactivity. The research aims to identify these shortcomings and address them.

- **Map Evaluation and Comparison:** The study, provides a detailed evaluation of 5 well-known and unknown web weather maps - DarkSky, Windy, In-Počasí, YR.no and Wundermap. Each map is analyzed for its strengths and weaknesses based on several features and parameters, such as the map composition, its symbology, its interactivity and its content. After this analysis, the problems that a cartographer may face are highlighted, but the best practices are highlighted.
- **Development of Cartographic Design:** The study, through the results of the experiment, provides strong recommendations for improving the design of web weather maps and beyond, synthesizing the results from the analysis of user behavior and interaction and the evaluation of the maps. The main emphasis is on user-centered design principles, always focusing on the simplification of the layouts, but also on the requirements and needs of different users.
- **Contribution to Cartographic Research:** In addition to practical applications, the study also aims to develop academic research on cartographic usability issues. This includes the integration of both qualitative data, through the Think-Aloud protocol, and quantitative data, through the recording of eye-tracking data. In this way, more comprehensive and evidence-based knowledge is created, on understanding user interaction, with web maps.

So, through all these objectives, the research directly addresses issues of difficulty in using maps. At the same time, it provides stimulus for future research involving the design and development of online weather maps. This study aims to develop these maps properly, i.e. to make them more effective and accessible to a wide range of users. Finally, by highlighting the importance of using these maps, it motivates future research on usability issues.

## 2.2 Methodology

The methodology applied by the authors of the article is quite innovative. The combination of quantitative and qualitative data to evaluate the usability of online weather maps was crucial. The design of this study, invoked 3 main elements: the selection of participants, the creation of the appropriate experimental setup and finally, the analysis of the results.

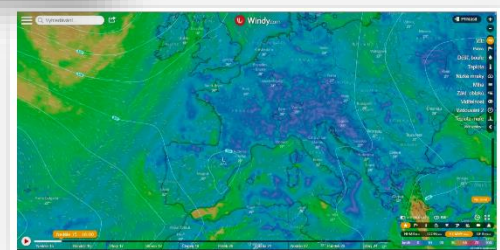
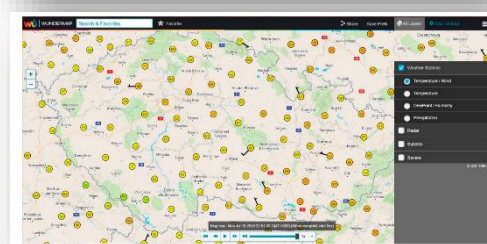
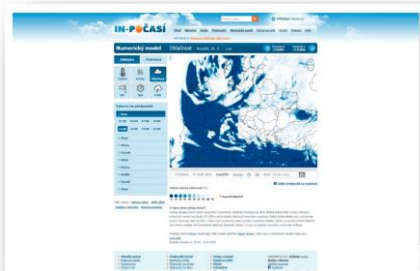
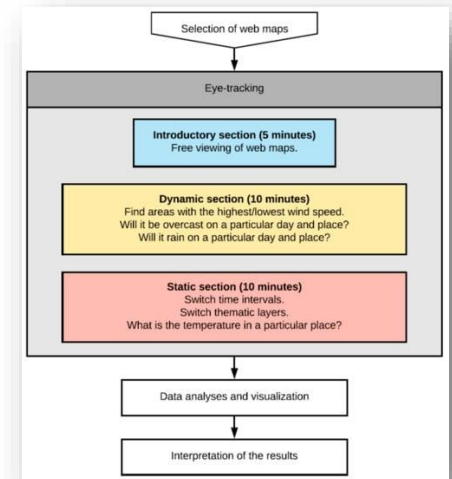
- **Participant Selection:** A total of 34 experienced and non-experienced individuals formed the target group of the experiment. This group, not restricted by age, literacy, employment or ethnicity. Specifically, the respondents were divided into 2 user groups: 16 novices and 18 experts, who were familiar with cartographic applications or geospatial



tools. This separation allowed the researchers to assess how experience affects user interaction and the usefulness of these maps.

- **Experimental Setup:** The primary stage was the selection of the maps where the experiment would take place. The researchers selected 5 web weather maps, where their use is intended for the general public. The maps used were Windy, In-Počasi, YR.no, DarkSky and Wundermap, with the first 3 being the most widely used and the other 2 less well known and in parallel, with different design approaches. As a second stage were the tasks performed. All participants were asked to complete 3 types of tasks:

1. **Introductory Section:** Viewing the selected web maps for each user, in order to familiarize the user with them.
2. **Dynamic Section:** Interactive interaction of the respondent with the maps to find specific features, such as finding rainfall forecasts or wind speed data.
3. **Static Section:** Evaluation of screenshots to analyze the usability of the map layout and captions.



- **Data Collection Tools:** Most of the study was based on eye-tracking technology, namely an SMI RED 250 eye-tracker with a 250 Hz operating frequency. There was also material support, from recordings and videos of the respondents. All of the above contributed to the recording of detailed data on user behavior. Measurements were made, such as the number of gaze fixations, duration, length of scan path and which were analyzed to measure attention, efficiency and navigation strategies. Finally, thinking protocols complemented the tracking data, providing additional qualitative information about participants' thoughts, preferences and frustrations during the tasks.
- **Data Analysis:**
  1. Quantitative Analysis: Continuous eye-tracking recordings provided measurable indicators, such as the time taken to identify specific information or the level of visual complexity.
  2. Qualitative Feedback: User comments from the post-experiment interviews were categorized to identify usability issues.

By integrating these methods, the study ensured a comprehensive evaluation of the usability of the maps, combining objective measurements with subjective user experiences. This multifaceted approach allowed the researchers to draw robust conclusions about the strengths and weaknesses of each weather web map, as well as to propose actionable design improvements.

## 2.3 Results

The results of the study by Popelka et al. give us a comprehensive answer to the function and usability of web weather maps. These findings are based on both quantitative results and qualitative results. At the same time, they are organized around three main themes: user behavior, the performance of different maps and the proper creation of these cartographic products.

- **User behavior**
  1. Novices vs. Experts: Through the study, significant differences were discovered in terms of user experience of cartographic products. The novices were mainly interested in the visual part of maps and were indifferent to the advanced functions that a map could offer. However, experts were more likely to explore interactive features such as display capabilities, thematic layers, etc.
  2. Navigation Patterns: The eye-tracking data showed that users often had problems identifying key elements, especially in cases where they were not correctly positioned or represented with poor symbolism. The analysis performed on the scan paths

showed that simpler layouts facilitated more efficient navigation.

- **Performance of Different Maps:**

1. DarkSky: This map, at first, seemed quite accessible to users, without them facing any serious problem. However, it lacked advanced features and a clear legend, which limited its usefulness for experts.
2. Windy: Although visually appealing and feature-rich, it was unsuitable for most users due to its cluttered interface and plethora of options.
3. In-Počasí: This map was found to be the most user-friendly, with clear illustrations and intuitive understanding. It was the most appropriate map with the right structure and content and, according to both beginners and experts, it was easy enough to navigate.
4. YR.no: There was a balanced cartographic product that also contained accurate data. However, regional inconsistencies in weather forecast caused frustration for users.
5. Wundermap: The work that had to be done on this map was extremely problematic. There was a great deal of difficulty in reading and poorly organized information. It also had very slow loading which made it very difficult for users.

- **Usability Challenges**

1. Legend Clarity: Participants often found it difficult to interpret the captions correctly. They were usually quite complex and in other cases were not clear, so that the user could not interact with them.
2. Interface Overload: Some maps contained excessive visual elements, as well as interactive options. This often made it difficult for users to understand the map with ease, thus reducing their efficiency of use.
3. Responsiveness: Especially for the Wundermap, there were too many difficulties regarding the slow loading times and the difficulty in interactivity with the user.

These results highlight the importance of simplicity, clarity and responsiveness in the design of web maps. Through this analysis, it highlights how important it is for cartographers to consider many features to properly create a map. One of the most important is the target audience of the cartographic product, but also the graphic design as modern visualization enhances the attractiveness and visual appeal of the user. Finally, areas where improvements are needed to enhance user satisfaction and performance are identified.

**Table 2.** Responses to questions and tasks given by the respondents in the dynamic and static sections of the test. Green indicates correct answers, red is incorrect answers, and bold plus exclamation mark indicates answers where a small amount of assistance was required.

| Participant                                                                                           | Dynamic part |       |          |       |           |         |       |          |       |           | Static Part |       |          |       |           |                                                                                                  |       |          |       |           |         |       |          |       |           |         |       |          |       |           |  |
|-------------------------------------------------------------------------------------------------------|--------------|-------|----------|-------|-----------|---------|-------|----------|-------|-----------|-------------|-------|----------|-------|-----------|--------------------------------------------------------------------------------------------------|-------|----------|-------|-----------|---------|-------|----------|-------|-----------|---------|-------|----------|-------|-----------|--|
|                                                                                                       | Task 1       |       |          |       |           | Task 2  |       |          |       |           | Task 3      |       |          |       |           | Task 1                                                                                           |       |          |       |           | Task 2  |       |          |       |           | Task 3  |       |          |       |           |  |
|                                                                                                       | Darksky      | Windy | Imposcal | YR.no | Wundermap | Darksky | Windy | Imposcal | YR.no | Wundermap | Darksky     | Windy | Imposcal | YR.no | Wundermap | Darksky                                                                                          | Windy | Imposcal | YR.no | Wundermap | Darksky | Windy | Imposcal | YR.no | Wundermap | Darksky | Windy | Imposcal | YR.no | Wundermap |  |
| P01                                                                                                   | OK           | OK    | OK       | OK    | X         | YES     | YES   | YES      | YES   | YES       | NO          | YES   | NO       | NO    | NO        | OK                                                                                               | OK    | OK       | OK    | X         | OK      | OK    | OK       | OK    | OK        | NA      | NA    | -8       | -3    | 21        |  |
| P02                                                                                                   | OK           | OK    | OK       | OK    | X         | YES     | YES   | NO       | YES   | X         | NO          | NO    | NO       | YES   | X         | OK                                                                                               | OK    | OK       | OK    | X         | OK      | OK    | OK       | OK    | OK        | 16      | -9    | -6       | -4    | 16        |  |
| P03                                                                                                   | OK           | X     | OK       | X     | X         | YES     | YES   | NO       | YES   | X         | NO          | NO    | NO       | YES   | X         | OK                                                                                               | OK    | OK       | OK    | OK        | OK      | OK    | OK       | OK    | OK        | 18      | -9    | -6       | -4    | 20        |  |
| P04                                                                                                   | OK           | X     | OK       | OK    | X         | YES     | YES   | YES      | YES   | YES       | NO          | NO    | NO       | NO    | YES       | OK                                                                                               | OK    | OK       | OK    | OK        | OK      | OK    | OK       | OK    | OK        | 18      | -9    | -6       | -1    | 19        |  |
| P05                                                                                                   | X            | X     | X        | X     | OK        | YES     | YES   | YES      | YES   | X         | YES         | NO    | NO       | YES   | X         | X                                                                                                | OK    | OK       | OK    | X         | OK      | OK    | OK       | OK    | OK        | 21      | -9    | -6       | -3    | 16        |  |
| P06                                                                                                   | X            | OK    | X        | X     | X         | NO      | YES   | YES      | NO    | YES       | NO          | NO    | NO       | YES   | NO        | X                                                                                                | X     | OK       | OK    | X         | OK      | OK    | OK       | OK    | OK        | 20      | -9    | -30      | -5    | 20        |  |
| P07                                                                                                   | OK           | OK    | OK       | X     | X         | NO      | NO    | NO       | NO    | X         | NO          | NO    | YES      | YES   | X         | OK                                                                                               | OK    | X        | OK    | OK        | OK      | OK    | OK       | OK    | OK        | 18      | -9    | -6       | -3    | 20        |  |
| P08                                                                                                   | OK           | OK    | OK       | X     | X         | NO      | NO    | NO       | NO    | NO        | NO          | YES   | NO       | NO    | NO        | OK                                                                                               | OK    | X        | OK    | OK        | OK      | OK    | OK       | OK    | OK        | 18      | -11   | -15      | -2    | 19        |  |
| P09                                                                                                   | OK           | OK    | OK       | OK    | X         | YES!    | NO!   | NO       | NO    | X         | NO          | YES   | NO       | YES   | X         | X                                                                                                | OK    | OK       | OK    | OK        | OK      | OK    | OK       | OK    | OK        | 18      | -9    | -15      | -8    | 16        |  |
| P10                                                                                                   | X            | OK    | OK       | OK    | X         | YES     | NO    | NO       | YES   | X!        | NO          | YES   | YES      | YES   | NO        | OK                                                                                               | OK    | OK       | OK    | OK        | OK      | OK    | OK       | OK    | OK        | 22      | -9    | 8        | -3    | 20        |  |
| P11                                                                                                   | OK           | OK    | OK       | OK    | X         | NO      | NO    | NO       | NO    | X         | NO          | X     | YES      | YES   | X         | OK                                                                                               | OK    | X        | OK    | X         | OK      | OK    | OK       | OK    | OK        | 18      | -9    | -15      | -4    | 20        |  |
| P12                                                                                                   | X            | OK    | OK       | OK    | X         | NO      | NO    | NO       | NO    | X         | NO          | NO    | NO       | YES   | YES       | X                                                                                                | OK    | OK       | X     | OK        | OK      | OK    | OK       | OK    | OK        | 18      | -9    | -4       | -3    | 21        |  |
| P13                                                                                                   | OK           | X     | OK       | OK    | X         | YES     | YES   | NO       | NO    | NO!       | NO          | NO!   | YES      | NO!   | X         | OK                                                                                               | OK    | OK       | OK    | OK        | OK      | OK    | OK       | OK    | OK        | 21      | -9    | -6       | -3    | 21        |  |
| P14                                                                                                   | OK           | OK    | OK       | OK    | X         | YES     | YES   | NO       | NO    | X!        | NO          | YES   | YES      | NO    | NO        | OK                                                                                               | OK    | X        | OK    | X         | OK      | OK    | OK       | OK    | OK        | 18      | -9    | -4       | -4    | 21        |  |
| P15                                                                                                   | OK           | OK    | OK       | OK    | OK        | YES     | YES   | YES      | X     | X         | NO          | NO    | YES      | NO    | X         | OK                                                                                               | OK    | OK       | OK    | X         | OK      | OK    | OK       | OK    | OK        | 18      | -9    | -15      | -3    | 21        |  |
| P16                                                                                                   | OK           | OK    | OK       | OK    | OK        | YES     | NO    | YES      | NO    | X         | NO          | YES   | YES      | YES   | X         | OK                                                                                               | OK    | OK       | OK    | X         | OK      | OK    | OK       | OK    | OK        | 18      | -9    | -15      | -9    | 19        |  |
| P17                                                                                                   | OK           | X     | X        | OK    | X         | YES     | YES   | NO       | NO    | X         | NO          | NO    | NO       | NO    | X!        | OK                                                                                               | OK    | OK       | OK    | X         | OK      | OK    | OK       | OK    | OK        | 20      | -9    | -6       | -4    | 16        |  |
| P18                                                                                                   | OK           | OK    | OK       | OK    | X         | NO      | NO    | NO       | NO    | X!        | NO          | NO    | YES      | NO    | X         | OK                                                                                               | OK    | OK       | OK    | X         | OK      | OK    | OK       | OK    | OK        | 20      | -9    | -6       | -3    | 19        |  |
| P19                                                                                                   | OK           | OK    | X        | X     | X         | YES     | NO    | NO       | NO    | X         | NO          | NO    | YES      | YES   | X!        | OK                                                                                               | OK    | OK       | OK    | OK        | OK      | OK    | OK       | OK    | OK        | 22      | -9    | -4       | -4    | 20        |  |
| P20                                                                                                   | OK           | OK    | OK       | OK    | X         | YES     | NO    | YES      | YES   | NO        | YES         | NO    | NO       | YES   | X         | OK                                                                                               | OK    | OK       | OK    | OK        | OK      | OK    | OK       | OK    | OK        | 18      | -9    | -6       | NA    | 23        |  |
| P21                                                                                                   | OK           | OK    | OK       | OK    | OK        | NO      | NO    | YES      | YES   | X         | YES         | YES   | YES      | NO    | NO        | OK                                                                                               | OK    | X        | OK    | OK        | OK      | OK    | OK       | OK    | OK        | 18      | -9    | -8       | -4    | 18        |  |
| P22                                                                                                   | OK           | OK    | OK       | OK    | OK        | NO      | NO    | NO       | NO    | YES       | NO          | YES   | YES      | NO    | X         | OK                                                                                               | OK    | OK       | OK    | OK        | OK      | OK    | OK       | OK    | OK        | 18      | -9    | -6       | -4    | 19        |  |
| P23                                                                                                   | OK           | OK    | OK       | OK    | X         | YES     | YES   | YES      | YES   | YES       | YES         | NO    | YES      | X     | X         | OK                                                                                               | OK    | OK       | OK    | X         | OK      | OK    | OK       | OK    | OK        | 18      | -9    | -6       | -3    | 19        |  |
| P24                                                                                                   | OK           | OK    | OK       | OK    | X         | NO      | NO    | NO!      | YES   | X!        | X           | YES   | YES      | YES   | X!        | OK                                                                                               | OK    | OK       | OK    | X         | OK      | OK    | OK       | OK    | OK        | NA      | -9    | -6       | -3    | 16        |  |
| P25                                                                                                   | OK           | OK    | X        | OK    | OK        | YES     | NO    | X!       | YES   | X!        | YES         | YES   | YES      | YES   | X!        | OK                                                                                               | OK    | OK       | OK    | X         | OK      | OK    | OK       | OK    | OK        | 23      | -9    | -6       | -3    | 19        |  |
| P26                                                                                                   | OK           | OK    | OK       | OK    | X         | YES     | YES   | YES      | YES   | X!        | NO          | YES   | YES      | YES   | X!        | OK                                                                                               | OK    | X        | OK    | OK        | OK      | OK    | OK       | OK    | OK        | 2       | -9    | -15      | -3    | 20        |  |
| P27                                                                                                   | OK           | OK    | OK       | OK    | X         | YES     | YES   | YES      | NO    | X!        | NO          | YES   | YES      | NO    | X!        | OK                                                                                               | OK    | OK       | OK    | OK        | OK      | OK    | OK       | OK    | OK        | 21      | -9    | -6       | -4    | 16        |  |
| P28                                                                                                   | OK           | X     | X        | OK    | X         | NO      | NO    | YES      | YES!  | X!        | YES         | NO    | YES      | YES   | X!        | OK                                                                                               | OK    | X        | OK    | X         | OK      | OK    | OK       | OK    | X         | 16      | -9    | -8       | -3    | 16        |  |
| P29                                                                                                   | OK           | OK    | OK       | OK    | X         | NO      | NO    | YES      | NO!   | X         | NO          | NO    | NO       | NO    | X!        | OK                                                                                               | OK    | OK       | OK    | X         | OK      | OK    | OK       | OK    | OK        | 24      | -9    | -6       | -3    | 21        |  |
| P30                                                                                                   | OK           | X     | OK       | OK    | X         | YES!    | NO    | YES      | YES   | X         | NO          | YES   | YES      | YES   | X         | OK                                                                                               | OK    | OK       | OK    | X         | OK      | OK    | OK       | OK    | OK        | 18      | -9    | -4       | -3    | 16        |  |
| P31                                                                                                   | OK           | OK    | OK       | OK    | X         | YES!    | NO!   | YES!     | NO!   | X!        | NO!         | NO!   | NO!      | YES!  | X!        | OK                                                                                               | X     | OK       | OK    | X         | OK      | OK    | OK       | OK    | OK        | 21      | -9    | -6       | 0     | 21        |  |
| P32                                                                                                   | OK           | OK    | OK       | OK    | X         | NO      | YES   | NO       | YES   | X!        | YES         | YES   | NO       | YES   | NO!       | OK                                                                                               | OK    | OK       | OK    | X         | OK      | OK    | OK       | OK    | OK        | 21      | -9    | -8       | -4    | 16        |  |
| P33                                                                                                   | OK           | OK    | OK       | OK    | OK        | NO      | YES   | X!       | NO    | X!        | NO          | NO    | YES      | YES   | X!        | OK                                                                                               | OK    | OK       | OK    | X         | OK      | OK    | OK       | OK    | OK        | 18      | -9    | -15      | -3    | 16        |  |
| P34                                                                                                   | OK           | X     | OK       | OK    | X         | YES     | NO    | YES      | NO    | X!        | NO          | YES   | YES      | YES   | X!        | X                                                                                                | OK    | OK       | OK    | X         | OK      | OK    | OK       | OK    | OK        | 23      | -9    | -6       | -3    | 18        |  |
| Correct                                                                                               | 30           | 26    | 28       | 29    | 6         | 34      | 34    | 32       | 32    | 9         | 33          | 33    | 34       | 33    | 8         | 30                                                                                               | 33    | 27       | 34    | 14        | 34      | 34    | 34       | 34    | 33        | 16      | 32    | 17       | 27    | 0         |  |
| Wrong                                                                                                 | 4            | 8     | 6        | 5     | 28        | 0       | 0     | 2        | 2     | 25        | 1           | 1     | 0        | 1     | 26        | 4                                                                                                | 1     | 7        | 0     | 20        | 0       | 0     | 0        | 0     | 1         | 18      | 2     | 17       | 7     | 34        |  |
| LEGEND                                                                                                |              |       |          |       |           |         |       |          |       |           |             |       |          |       |           |                                                                                                  |       |          |       |           |         |       |          |       |           |         |       |          |       |           |  |
|  Correct answer    |              |       |          |       |           |         |       |          |       |           |             |       |          |       |           |  Wrong answer |       |          |       |           |         |       |          |       |           |         |       |          |       |           |  |
|  Assistance needed |              |       |          |       |           |         |       |          |       |           |             |       |          |       |           |                                                                                                  |       |          |       |           |         |       |          |       |           |         |       |          |       |           |  |

LEGEND Correct answer Wrong answer Bold Assistance needed

## 2.4 Recommendations

Based on the findings of the study, Popelka et al. give several options and make various observations, which address the usability issues identified, as well as give recommendations regarding the overall functionality of web weather maps. The various observations and recommendations made are mainly focused on the balance between making things simple and accessible while providing advanced functionalities that could serve a wide audience.

- **Simplify Layouts:** Researchers stress the importance of simplifying graphic products. Simplicity in design allows the user to focus on critical information without being distracted by excessive visual elements or interactive options. At the same time, it triggers the logical grouping of related features and the provision of intuitive navigation structures. This can help the user to locate information more effectively.
- **Enhance Legend Usability:** Legends should be clear, concise and prominently displayed. Based on the study, it is recommended to use consistent symbols and colors at different map levels. In this way, any confusion will be avoided. It is also recommended to use interactive



captions that highlight specific elements to the user. This is done by interacting with each other and can further improve the user's understanding of the cartographic product and the information it provides.

- **Improve Responsiveness:** Faster loading times and smoother interactions are vital for maintaining user engagement. Optimizing backend processes and ensuring that map elements load dynamically without delays can significantly enhance the user experience.
- **Offer Customization Options:** The cartographic product must be produced in such a way that it is adaptable to a wide audience. Novices can benefit from simplified views, while experts can access advanced features, such as detailed layer toggles and analytical tools.
- **Conduct Regular User Testing:** Iterative testing with a diverse group of users can identify new usability issues as technology and user expectations evolve. Eye-tracking and think-aloud protocols remain effective tools for understanding user behavior and refining designs.

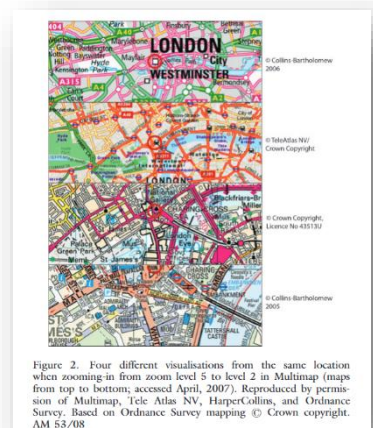
By applying such observations, cartographers can create convenient web weather maps. In this respect, it is not necessary for cartography products to focus merely on their appeal, but rather on functional and user-friendly interfaces. This way, web maps will continue to be helpful for ordinary users in everyday life and expert professionals alike by supporting better decision-making in the digital era.

### 3. Literature Review

Literature review is the most important and essential part of any research work. It encompasses a wide array of knowledge in the field while showing the shortcomings that have yet to be examined further. In the context of the usability of web maps, methodologies for eye tracking and design principles in cartography, the contents researched are relevant to this study. Therefore, this section categorizes and summarizes the major studies that have been done regarding the subject pieces, highlighting the relevance to this study.

#### 3.1 Web Map Usability Studies

The usability of web maps has been one of the key elements of their design for years. Early research, such as Harrower et al. (1997), attempted to identify differences in the use of web maps between geographers and non-geographers. At the same time, studies such as those conducted by Nivala et al. (2008), managed to emphasize simplicity and clarity in user interfaces. Their study revealed that overly complex layouts could hinder usability and reduce user satisfaction. They suggested that harmonization



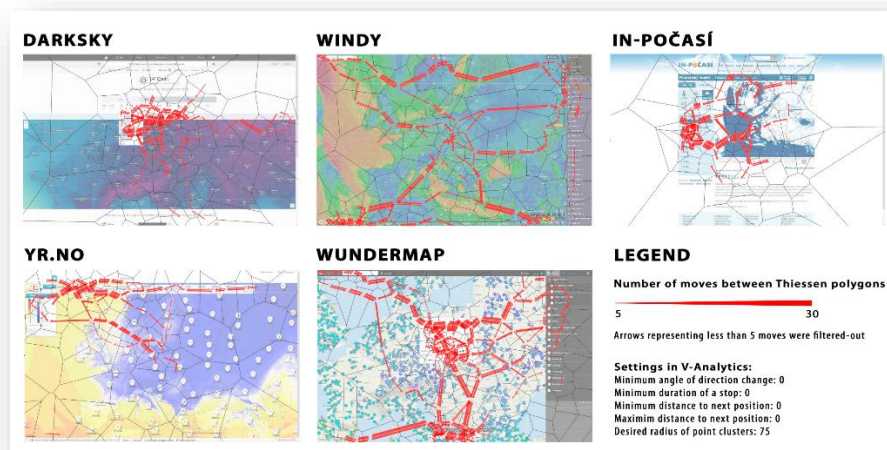


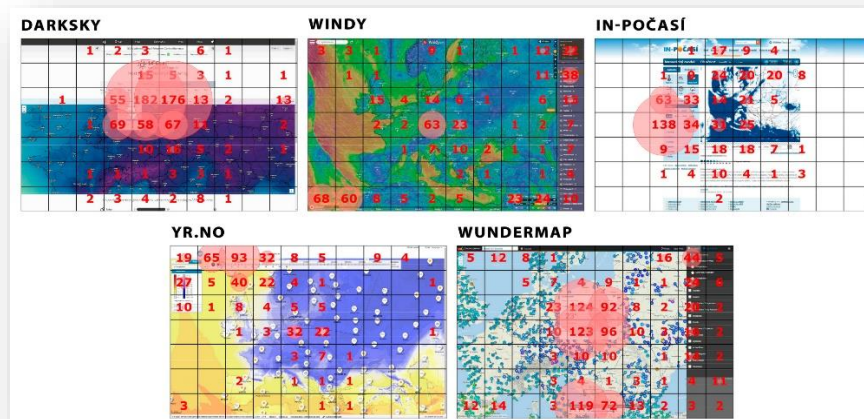
of maps was a better solution. Similarly, Harrower and Brewer (2003) discussed the balance between aesthetic design and functional clarity, advocating user-centered approaches to cartographic design. Over the years, recent studies have extended the discoveries of previous years, while also expanding the role of interactivity. Roth et al. (2015) sought to answer the question of how interactive features such as zoom functions affect the user experience. The results of their research showed that while interactivity enhances engagement, it can also introduce a range of knowledge if not implemented correctly. This is in line with the results of Popelka et al. who identified similar issues in the web weather maps they had under review.

### 3.2 Eye-Tracking in Cartography

Eye-tracking has become an essential tool for evaluating user interaction with maps. MacEachren et al. (1998) were among the first to apply eye-tracking in cartography, using it to study how users interpret map symbols and layouts. In this way, they facilitated their visual thinking and the observation of patterns. Their research established a foundation for understanding visual attention in map reading.

More recent studies, such as those by Popelka (2019), have refined these methods by combining eye-tracking with qualitative data, such as think-aloud protocols. This approach provides a comprehensive view of user behavior, allowing researchers to identify specific areas of confusion or inefficiency. For example, Popelka's analysis of scanpath patterns revealed that users often struggle with cluttered interfaces and poorly designed legends, findings that are consistent with the results of the selected study.





### 3.3 Design Principles in Cartography

The design principles are very instrumental in making web maps highly usable. Brewer's (2005) principles of color use in cartography showed a need for accessibility and consistency; these have been highly integrated into web map design in the course of time to assist in making the maps highly readable and understandable by the users.

Peterson (2003) examined the effects of animation and dynamic elements on user experience. Although these features can make maps more engaging, Peterson cautioned that they must be carefully designed to avoid overwhelming users. This is especially pertinent to the findings of Popelka et al., who found that too much interactivity in some weather web maps caused cognitive overload.

### 3.4 Identified Gaps and Future Directions

Although much progress has been made in understanding the usability of web maps, there remain several gaps and problems that need further investigation:

- **Demographic Diversity:** Most studies of map use are based mainly on middle-aged people. This is intended to create a narrow demographic group, often excluding the elderly and especially people with disabilities. Future research should emphasize a wider range of participants to ensure accessibility and inclusiveness of marginalized people in the design of web maps.
- **Longitudinal Studies:** The surveys that are usually carried out are short-term surveys. Their purpose is not to fatigue users where they will be asked to do various tasks. However, longitudinal study would be important as it could track how users adapt to the web maps with the ultimate goal of providing deeper insights into the learning and usage factors of the maps.
- **Integration of Emerging Technologies:** With the developments in the field of technology and especially in recent years in the field of

artificial intelligence and machine learning, research could be done on the field of cartographic production. If it were possible to create adaptive user-based web maps, it would be an appropriate solution to the problems mentioned earlier.

#### 4. Discussion

Integration of insights from the study of Popelka et al. with wider trends in literature shows great enhancements and persistent pain points in web map usability. The web maps have evolved over time from static to dynamic and interactive platforms. These enhancements, while enabling better functionality, also introduce cognitive burdens, as indicated by Peterson in 2003, which the findings on cognitive overload by Popelka et al. slightly support. Methods such as eye-tracking and think-aloud protocols have become integral to the understanding of user interactions, showing growth in reliance on uncovering hidden usability issues with these techniques in refining designs. Simplicity is taken as the utmost key to map design. Both the literature and the study chosen clearly to point out the need to reduce visual clutter and hence enhance intuitive layouts. Brewer (2005) has pointed out that clear visual hierarchies assist in guiding the user; this is well complemented by the finding of Popelka et al. that poorly designed legends and controls detract from usability. While interactivity enhances user engagement, a high number of options-as in the case of Windy's interface-may overwhelm users. The balance between simplicity and functionality is a crucial accommodation for both novice and experienced users.

The combination of the quantitative and qualitative methodologies provides a strong framework for usability analysis. Metrics, such as fixation counts and scan path lengths, will quantify user behavior by offering precise insight into attention allocation. Popelka et al. apply these metrics when analyzing navigation and information retrieval inefficiencies, while think-aloud protocols are applied to capture the subjective experiences of users, thus deepening the quantitative data. Integration of both methods will enable the researcher to triangulate data, hence completeness in understanding users' behavior and preference. In optimizing web maps for different kinds of audiences, varied expertise and preference levels have to be considered. Beginners like simplified interfaces with a clear legend and few options, as can be testified by the intuitive design of In-Poçasí. With features such as adding one's own layer, the possibility of doing advanced analysis meets professional needs-even the more complex features make Windy very valuable to an advanced user. The fact that the design was taking into consideration either visual impairment or low technical skill extends the usefulness of this service across all kinds of demographics: interactive legends, responsive layouts enable it effectively.

Proposed design improvements: making interfaces adaptive per user preference or expertise, enhancement in usability toggling between "basic" and "advanced." Optimized load times are very important; a slow performance reduces the user

experience of Wundermap. Optimization at the backend is very important with much-needed efficient loading mechanisms. Finally, improved feedback-real-time feedback with highlighted selected layers or interactive tooltips that intuitively guide users across complex interfaces.

## 5. Conclusions

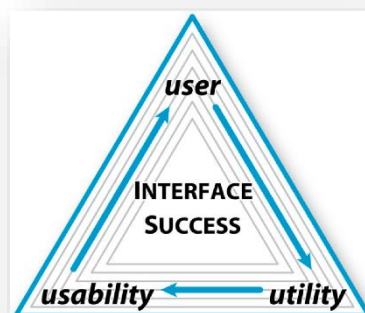
The following paper emphasizes the most important usability dependence of web maps on user-centered design. When crafting effective maps, the concentration must be clarified: too many visuals and complicated interactive components have the capability of overwhelming users. The users' needs, including their experience level, need to be acknowledged. Personalized design approaches ensure that these maps are appropriate both for novices and professionals of all users.

It is useful to combine quantitative methods, such as eye tracking, with qualitative methods, such as verbal reasoning, to understand how people interact with maps in order to get meaningful insights from usability challenges. This is done by giving options for simple, static views and more complex interactive features, thus enabling users to select an interface most suitable for their skill and preference.

Another important aspect is the improvement of loading speed and responsiveness of the maps. Faster loading will prevent users from leaving your page when the map takes too long to load. A balance between ease of use and inclusion of advanced features needs to be found. This will accommodate a wide range of users with different needs and abilities.

Longitudinal studies can show how users adapt to web maps over time. They can also denote long-term trends in usability and how users develop their skills as their familiarity with the maps increases. Research on cultural differences in map use contributes to the creation of accessible designs for people with a variety of abilities.

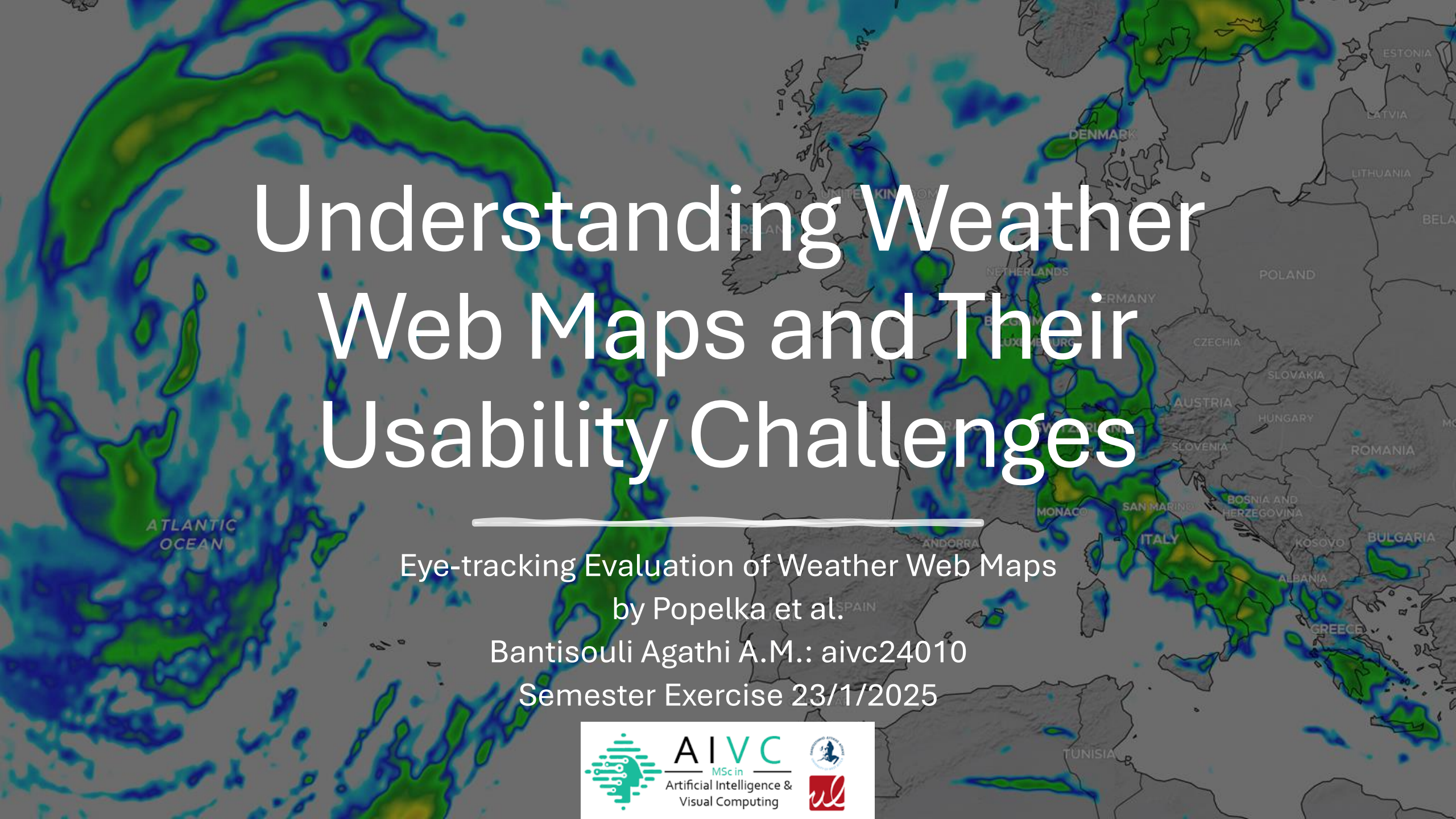
Dynamic interfaces may provide a key to future developments in map usability. The use of machine learning allows such interfaces to adapt and, hence, be more responsive to users' needs. This could considerably enhance the user experience by making maps that keep getting better based on how people use them.



## 6. References:

- Popelka, S., Vondrakova, A., & Hujnakova, P. (2019). Eye-tracking Evaluation of Weather Web Maps. *ISPRS International Journal of Geo-Information*, 8(6), 256.
- Harrower, M.; Keller, C.P.; Hocking, D. (1997). Cartography on the Internet: Thoughts and a preliminary user survey. *Cartogr. Perspect.*, 27–37.
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- Roth, R. E., et al. (2015). User-centered design for interactive maps. *Cartographic Perspectives*.





# Understanding Weather Web Maps and Their Usability Challenges

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Eye-tracking Evaluation of Weather Web Maps  
by Popelka et al.

Bantisouli Agathi A.M.: aivc24010

Semester Exercise 23/1/2025

# Purpose and Relevance of Weather Web Maps

- **Importance of Weather Web Maps:**

1. Provide real-time weather updates for daily use.
2. Serve professionals in geoinformatics, cartography and related fields.



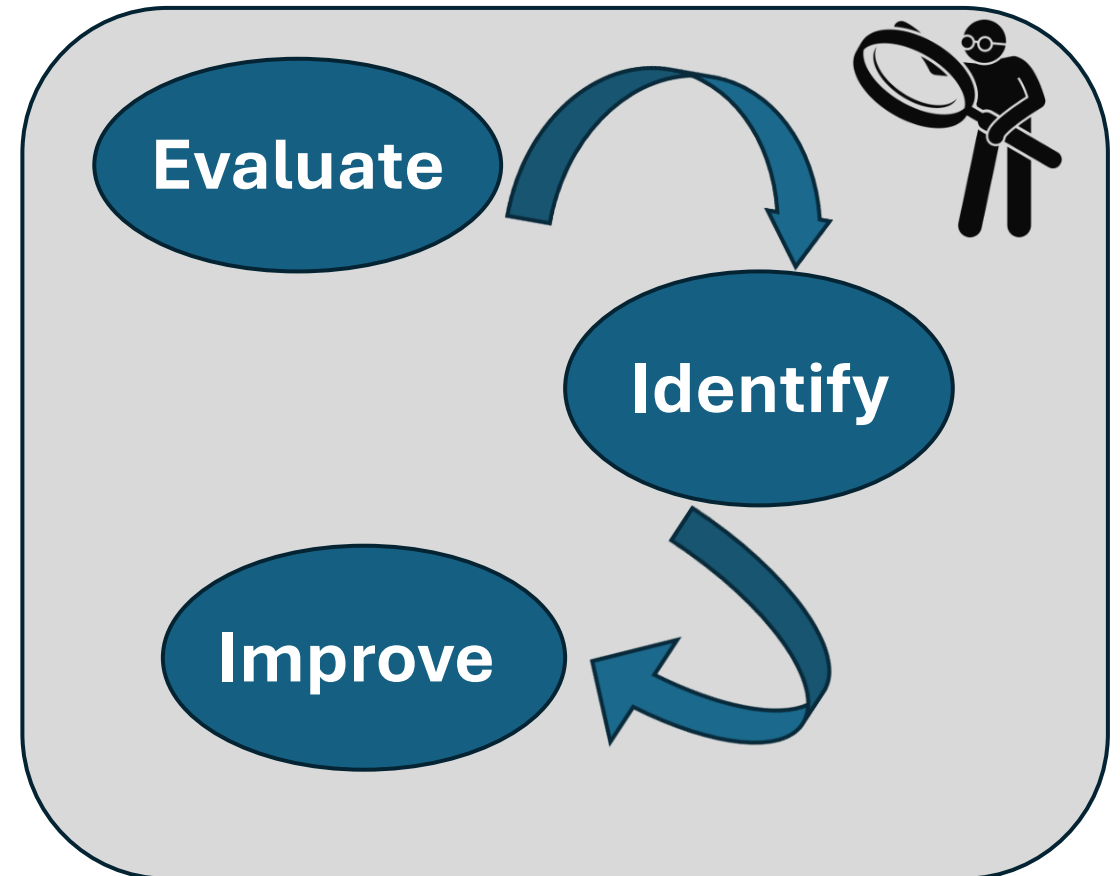
- **Challenges:**

1. Increasing complexity in map interfaces.
2. Usability issues such as cluttered layouts and unclear legends.



- **Research Objectives:**

1. Evaluate the usability of five popular weather web maps.
2. Use eye-tracking and think-aloud protocols to identify key issues and recommend improvements.



# Methodology Overview



## Approach:

**Quantitative:** Eye-tracking metrics (fixation count, scan path length, trial duration).

**Qualitative:** Think-aloud protocols capturing user thoughts and feedback in real-time.



## Tools Used:

**Eye-Tracking Hardware:** High-resolution eye-tracking devices (e.g., SMI RED 250).

**Software:** Tools for recording and analyzing eye movement data.

**Observation Protocols:** Detailed steps for conducting think-aloud sessions.

**User Interaction**

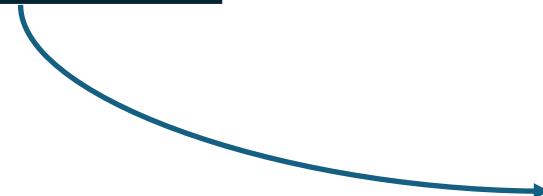


**Eye-Tracking**

**Think-Aloud**



**Data Analysis**





| Weather Web Map | Interface Simplicity            | Responsiveness                                    | Accessibility              |
|-----------------|---------------------------------|---------------------------------------------------|----------------------------|
| DarkSky         | Minimalist, very simple         | Fast loading time                                 | Limited features           |
| Windy           | Visually complex, many features | Can be slow for novice users                      | Overwhelming for new users |
| In-Pocasi       | Intuitive, user-friendly        | Quick response                                    | Easy to navigate           |
| YR.no           | Balanced design                 | Accurate data, but can be inconsistent regionally | Good for general use       |
| Wundermap       | Cluttered, confusing            | Slow loading                                      | Poor usability             |

## Evaluated Weather Web Maps

**DarkSky:** Known for its minimalist design but lacks advanced features.

**Windy:** Visually appealing and feature-rich but overwhelming for novices.

**In-Pocasi:** Intuitive design with user-friendly navigation.

**YR.no:** Balanced design and accurate weather data, but regional inconsistencies exist.

**Wundermap:** Poor usability due to confusing controls and slow loading times.



# User Groups and Their Interaction Patterns

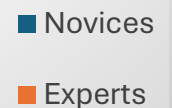
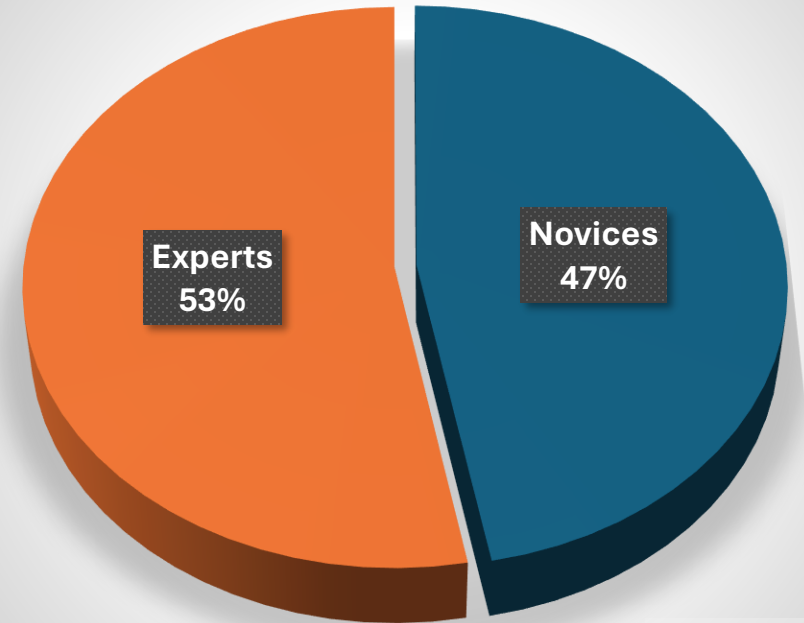
- **Sample Size:**

- Total participants: 34
- **Novices:** 16 (limited experience with web maps).
- **Experts:** 18 (familiar with cartographic tools and geoinformatics).

- **User Interaction Insights:**

- **Novices:** Struggled with advanced features, relied on visual cues and simplified layouts.
- **Experts:** Explored interactive features like toggling layers and analyzing data.

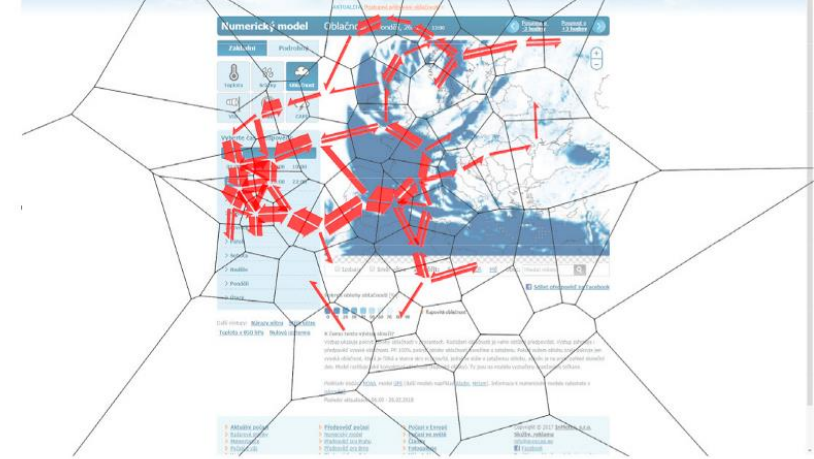
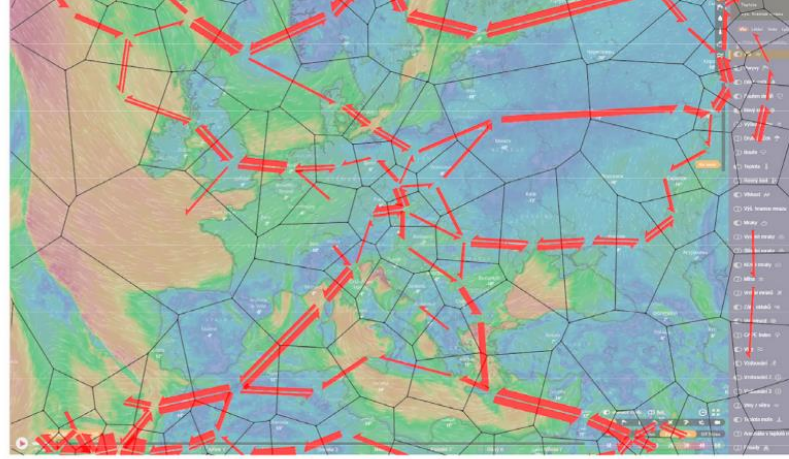
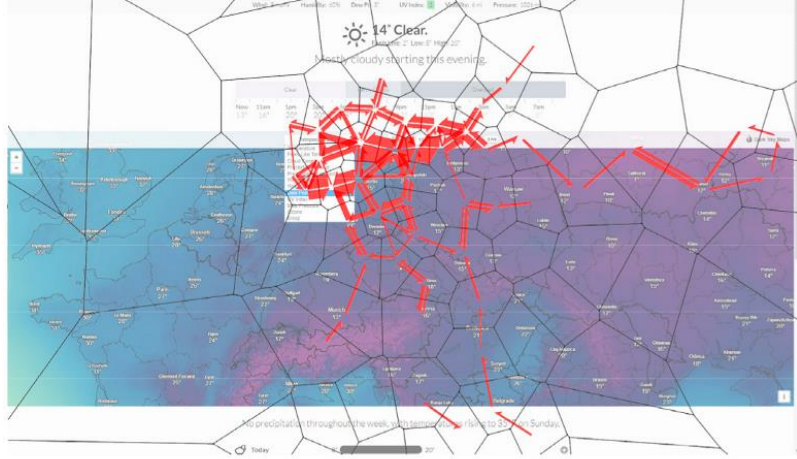
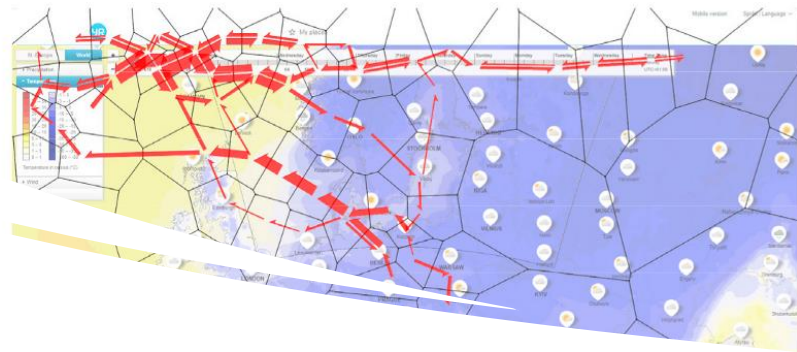
Novices vs. Experts



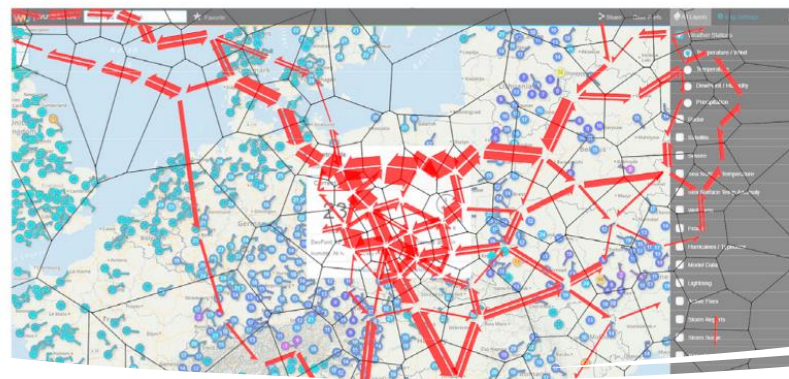


# Performance Comparison of Web Maps

| Web Map   | Legend Clarity | Navigation Ease | User Experience       | Loading Time | Usability Issues                |
|-----------|----------------|-----------------|-----------------------|--------------|---------------------------------|
| DarkSky   | Poor           | Easy            | Novices:<br>Satisfied | Fast         | Lack of advanced features       |
| Windy     | Moderate       | Complex         | Overwhelming          | Moderate     | Cluttered interface             |
| In-Počasí | Clear          | Easy            | Easy to navigate      | Fast         | None                            |
| YR.no     | Moderate       | Moderate        | Balanced              | Moderate     | Regional inconsistencies        |
| Wundermap | Poor           | Difficult       | Poor                  | Slow         | Slow loading, poor organization |

**YR.NO**

# WUNDERMAP



## LEGEND

### Number of moves between Thiessen polygons



Arrows representing less than 5 moves were filtered-out

### Settings in V-Analyst:

Minimum ~~single~~ or direct:

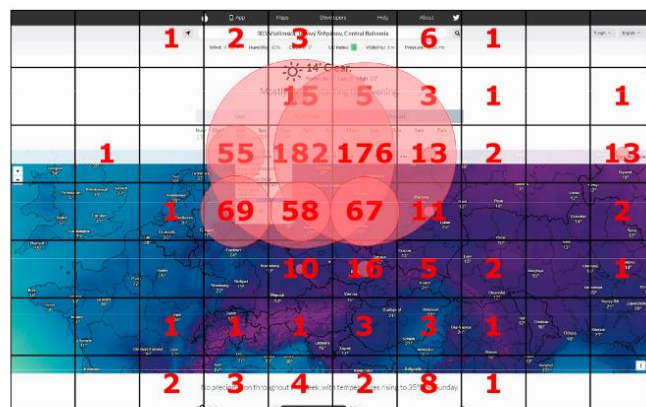
# Insights from Eye-Tracking Data

- **Heatmaps:** Display heatmaps showing areas of high focus and interaction on each web map.
- **Scanpaths:** Show the movement of the eye across the screen during interactions, illustrating the user's navigation patterns.

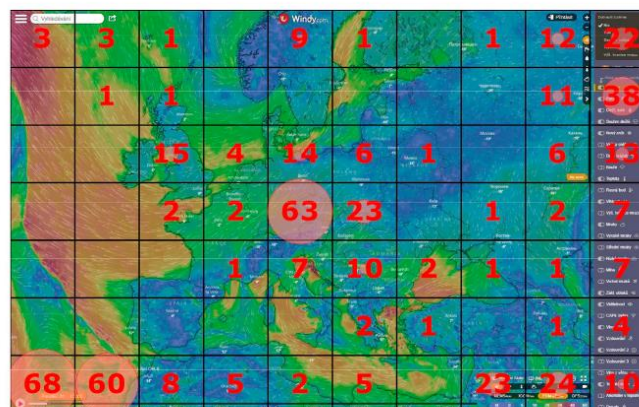


# Insights from Eye-Tracking Data

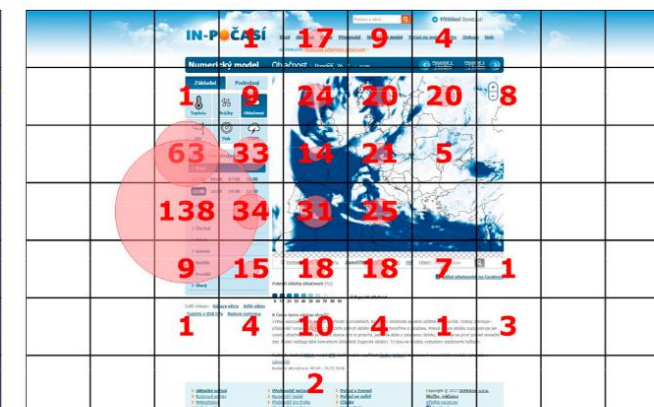
**DARKSKY**



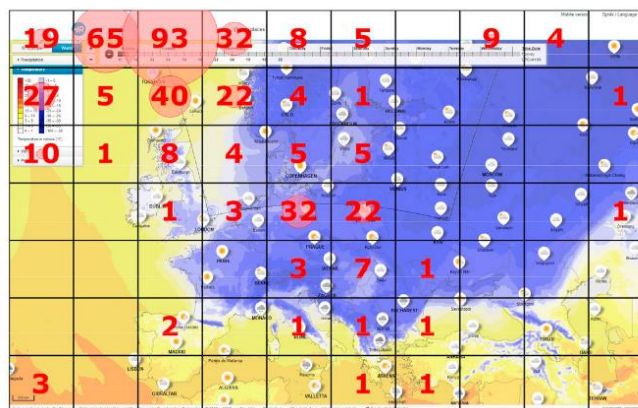
**WINDY**



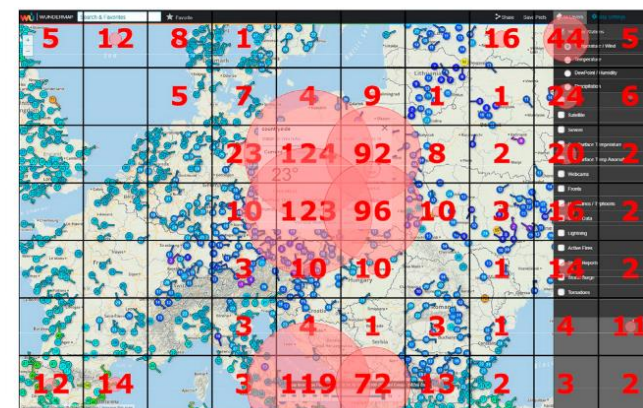
**IN-POČASÍ**



**YR.NO**



**WUNDERMAP**



# Usability Challenges Identified

Cluttered  
interfaces.

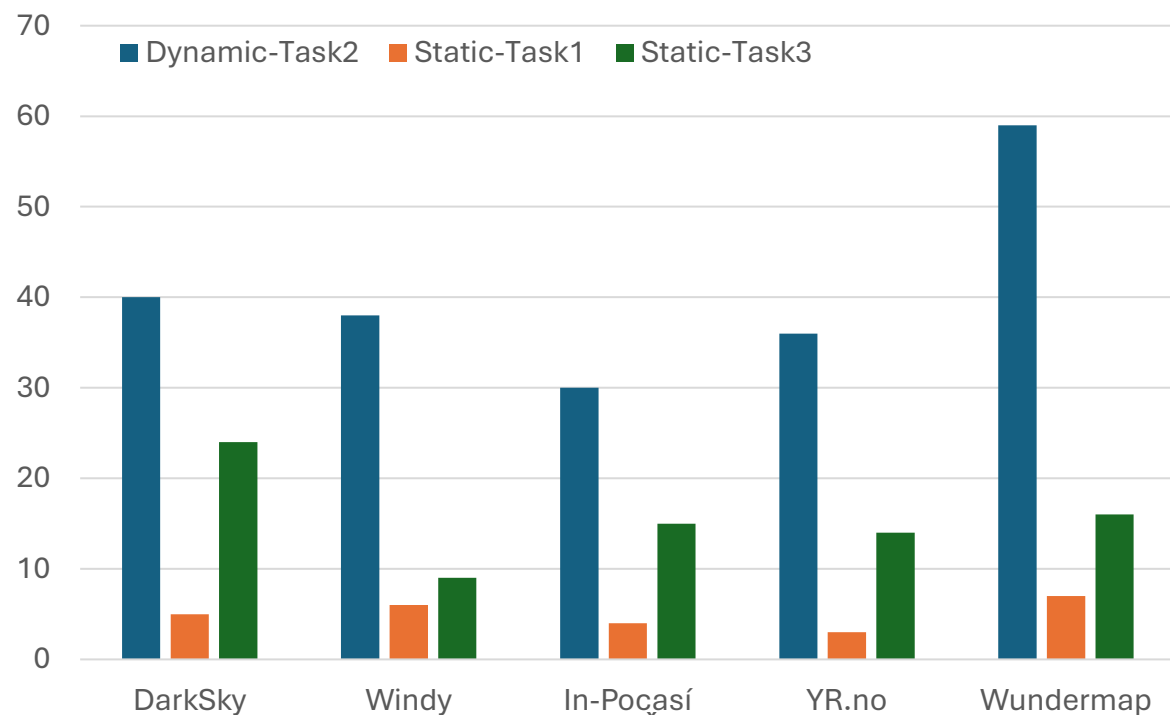
Unclear legends.

Slow or  
unresponsive  
behavior.

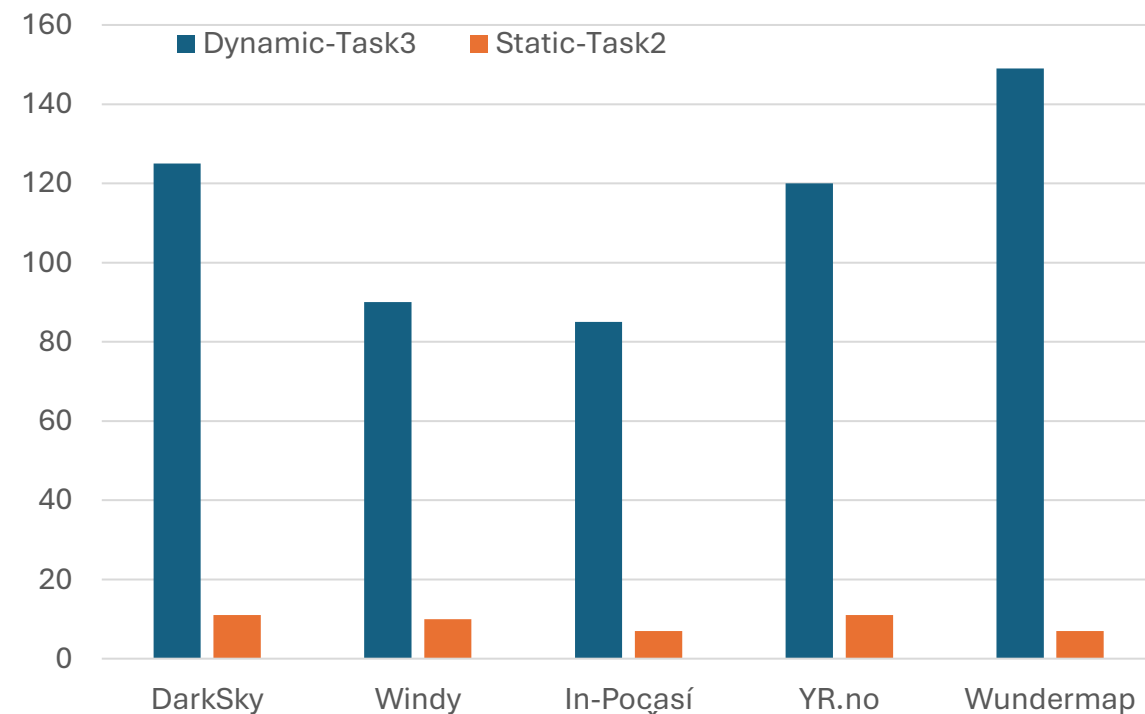
Complex  
navigation or  
overwhelming  
features.

# Visuals representation of key metrics

## Trial Duration



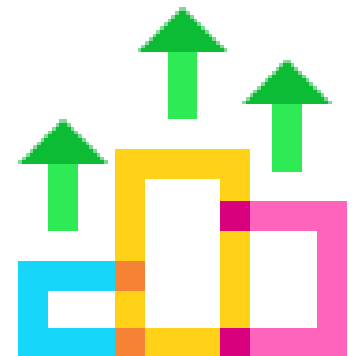
## Fixation Count



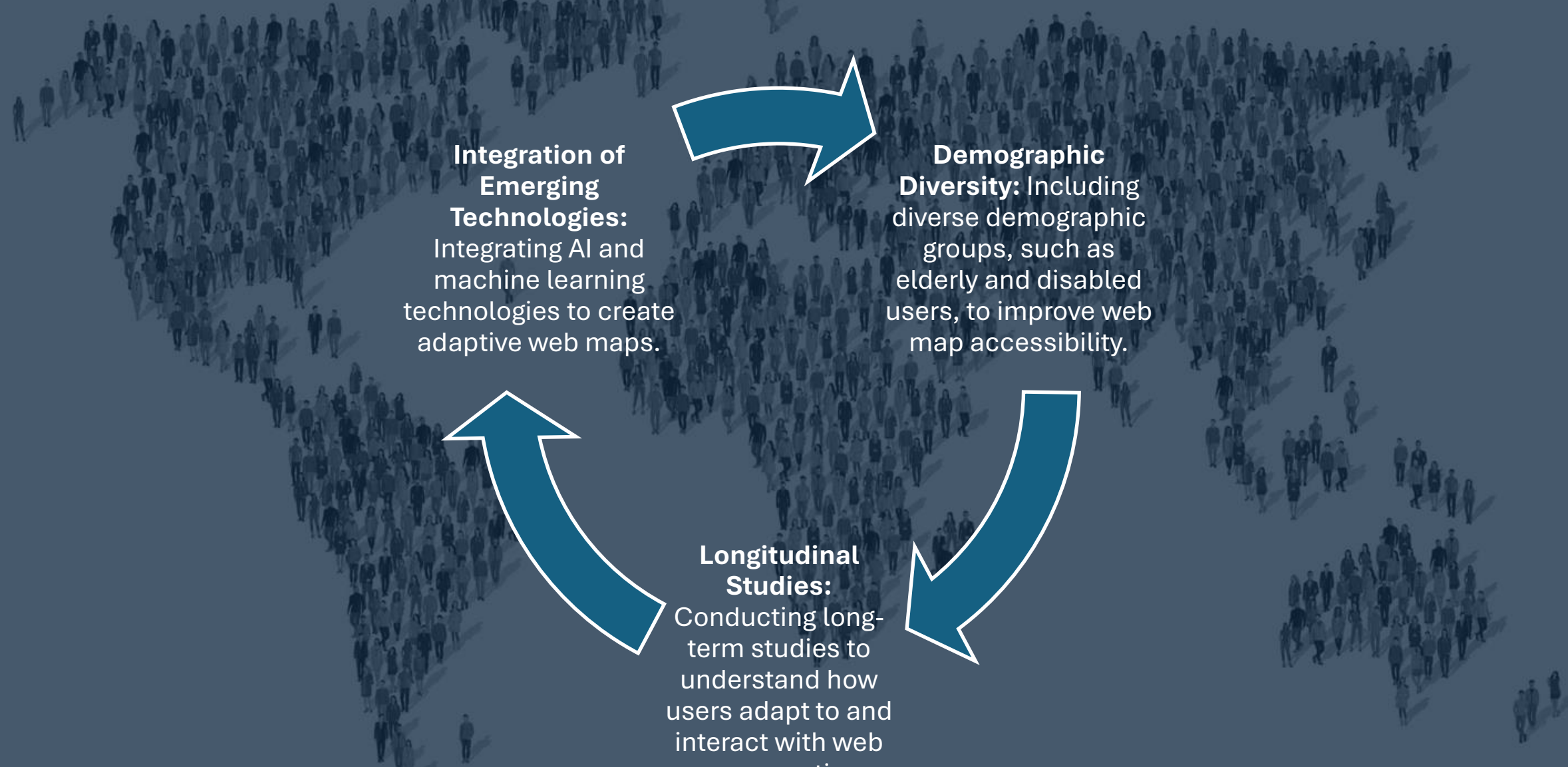


# User Interaction & Usability Improvements

| User Category | Interaction with Map                          | Usability Recommendations                  |
|---------------|-----------------------------------------------|--------------------------------------------|
| Novices       | Simple navigation, Focus on basic information | Simplified layout, Clearer legends         |
| Experts       | Use of advanced tools, Detailed data analysis | More advanced features, Tool customization |



# Future Research Directions



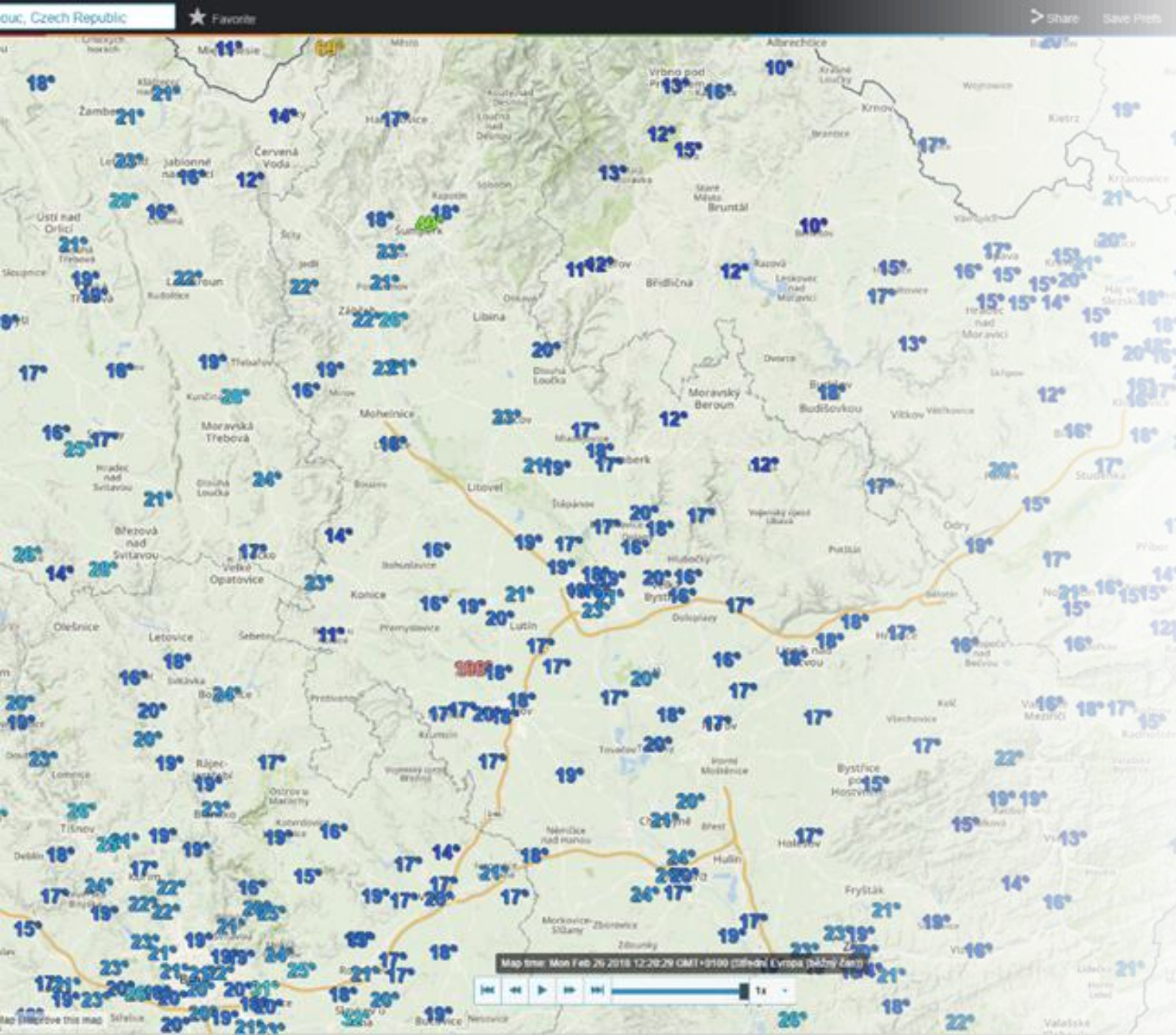
**Integration of Emerging Technologies:** Integrating AI and machine learning technologies to create adaptive web maps.

**Demographic Diversity:** Including diverse demographic groups, such as elderly and disabled users, to improve web map accessibility.

**Longitudinal Studies:** Conducting long-term studies to understand how users adapt to and interact with web maps over time.







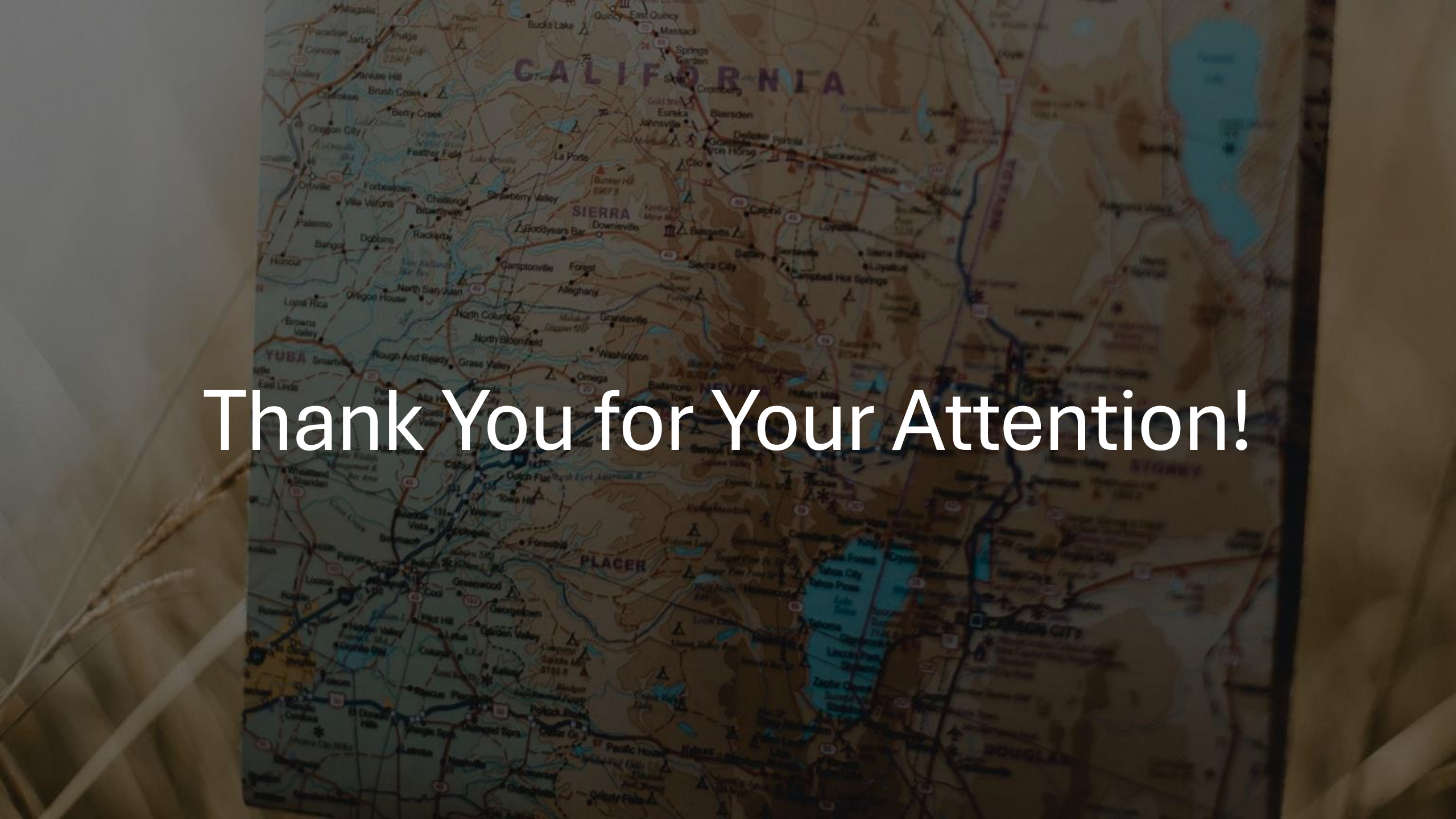
# Appendix – Supporting Visuals or Data





# References

- Popelka, S., Vondrakova, A., & Hujnakova, P. (2019). Eye-tracking Evaluation of Weather Web Maps. *ISPRS International Journal of Geo-Information*, 8(6), 256.
- Harrower, M.; Keller, C.P.; Hocking, D. (1997). Cartography on the Internet: Thoughts and a preliminary user survey. *Cartogr. Perspect.*, 27–37.
- Brewer, C. A. (2005). Guidelines for using color in cartography. *Journal of Cartographic Design*.
- Harrower, M., & Brewer, C. A. (2003). ColorBrewer.org: An online tool for selecting color schemes for maps. *The Cartographic Journal*.
- MacEachren, A. M., et al. (1998). Cartographic user interfaces. *Journal of Geographical Systems*.
- Nivala, A., et al. (2008). Usability of mobile maps. *Transactions in GIS*.
- Peterson, M. P. (2003). *Maps and the Internet*. Elsevier.
- Roth, R. E., et al. (2015). User-centered design for interactive maps. *Cartographic Perspectives*.

A vintage map of California, showing various counties and cities. The map is slightly faded and has a warm, brownish tint. Overlaid on the map is the text "Thank You for Your Attention!" in a large, white, sans-serif font. The text is centered horizontally and vertically. The map shows major cities like San Francisco, Sacramento, and Los Angeles, as well as smaller towns and geographical features like mountains and rivers. The word "CALIFORNIA" is printed in large, bold letters at the top of the map.

Thank You for Your Attention!